

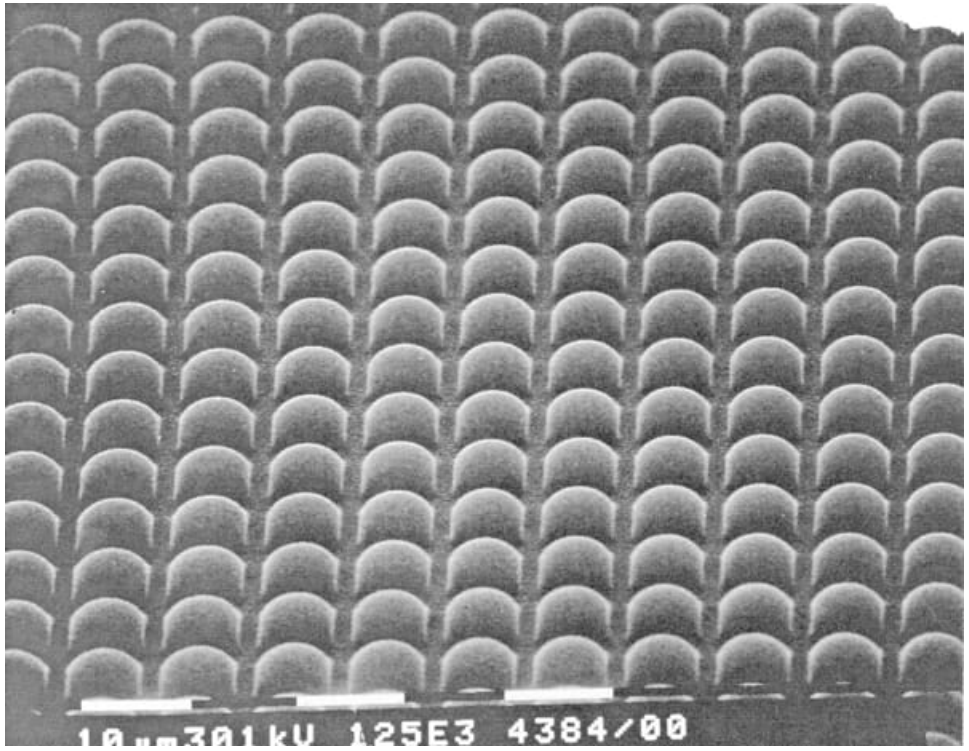
POLITECNICO
MILANO 1863

Image Sensors

SENSOR SYSTEMS

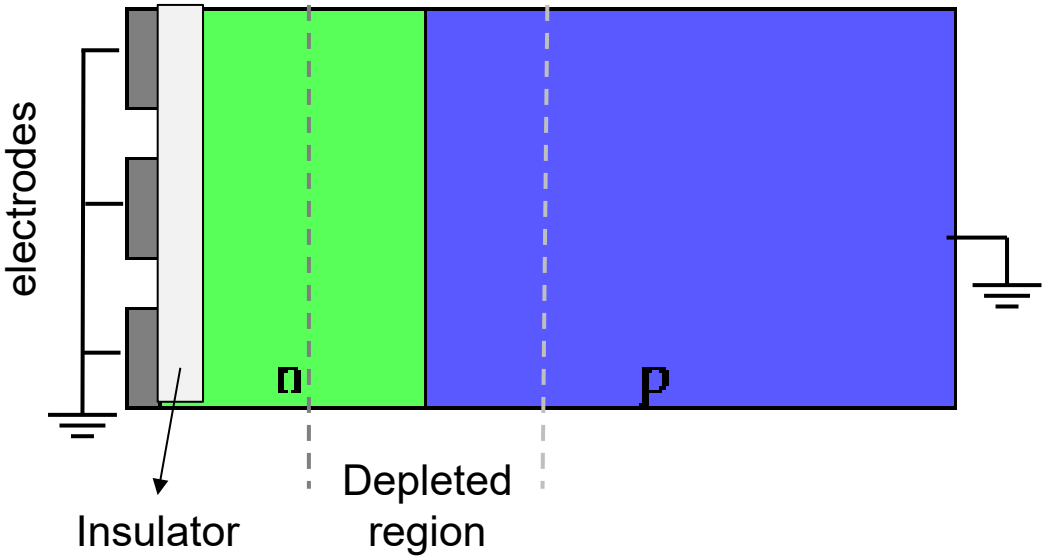
Federica Villa
federica.villa@polimi.it

- Charge Coupled Device (CCD)
- CMOS active pixel image sensor
- CCD vs. CMOS



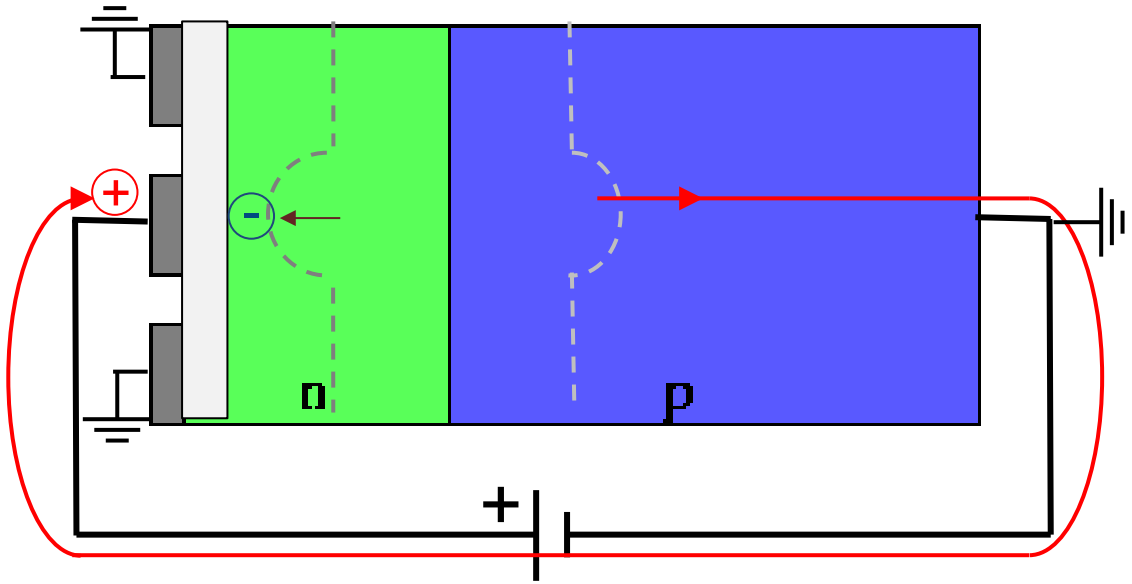
Charge Coupled Device (CCD)





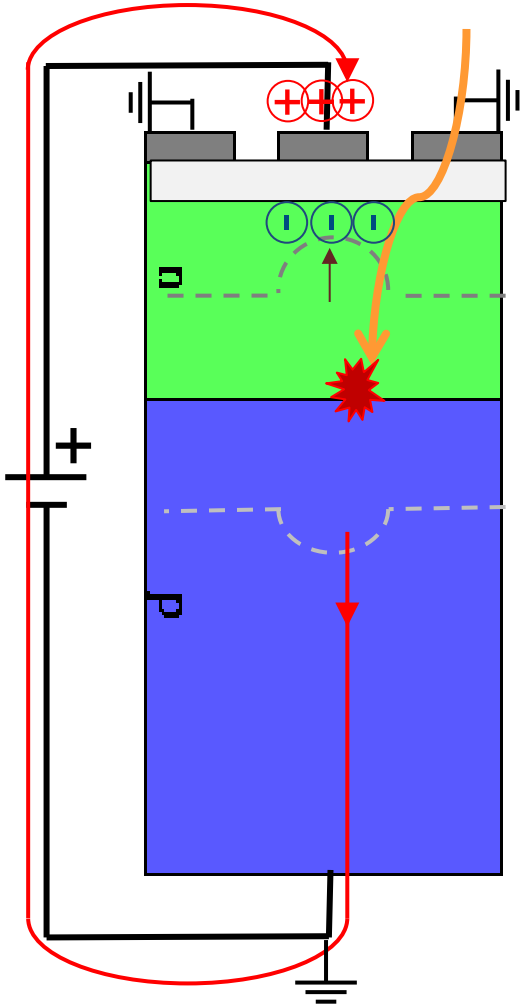
Unbiased

(no current flow
→ free carriers recombination)

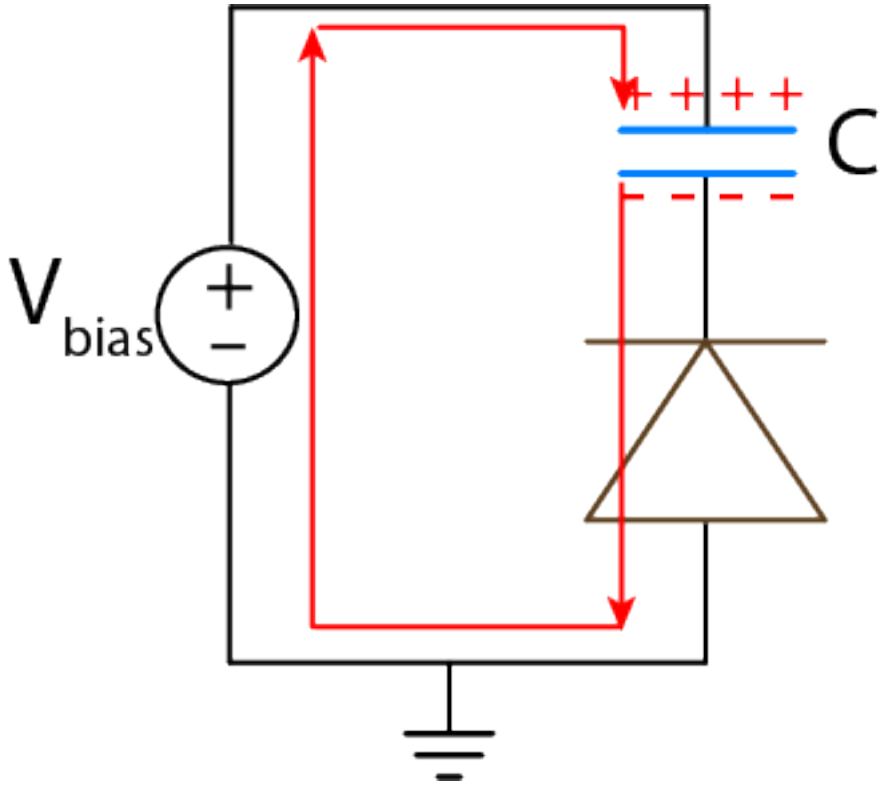


Biased

(free carriers generates a current
similarly to a photodiode)



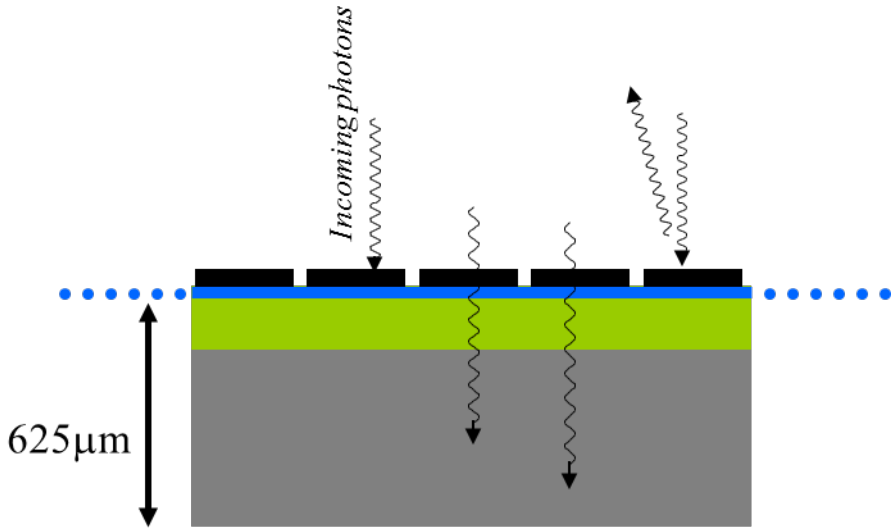
Equivalent model



When a photon is absorbed, it generates an electron-hole pair → photocurrent

Differently from photodiodes, electrons are stopped by the insulator at the n-doped side

Front side illuminated (Standard Wafer)



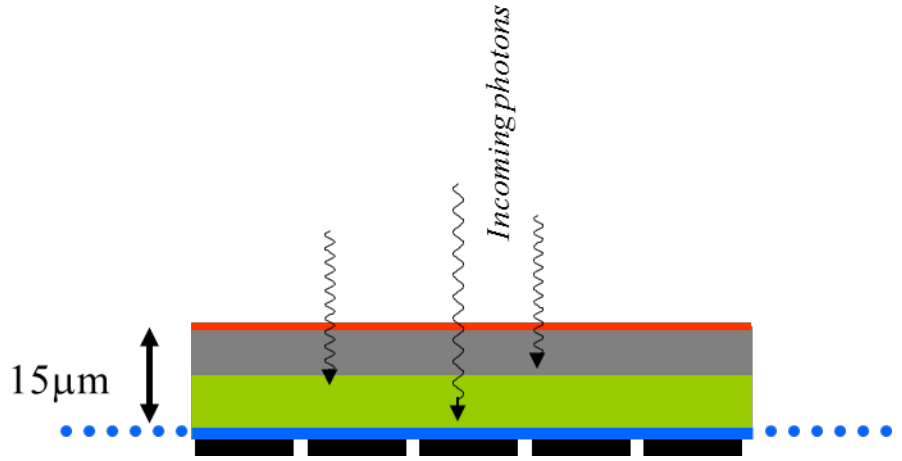
p-type silicon

n-type silicon

Silicon dioxide insulating layer

Polysilicon electrodes

Back side illuminated (Thinned Wafer)



Anti-reflective (AR) coating

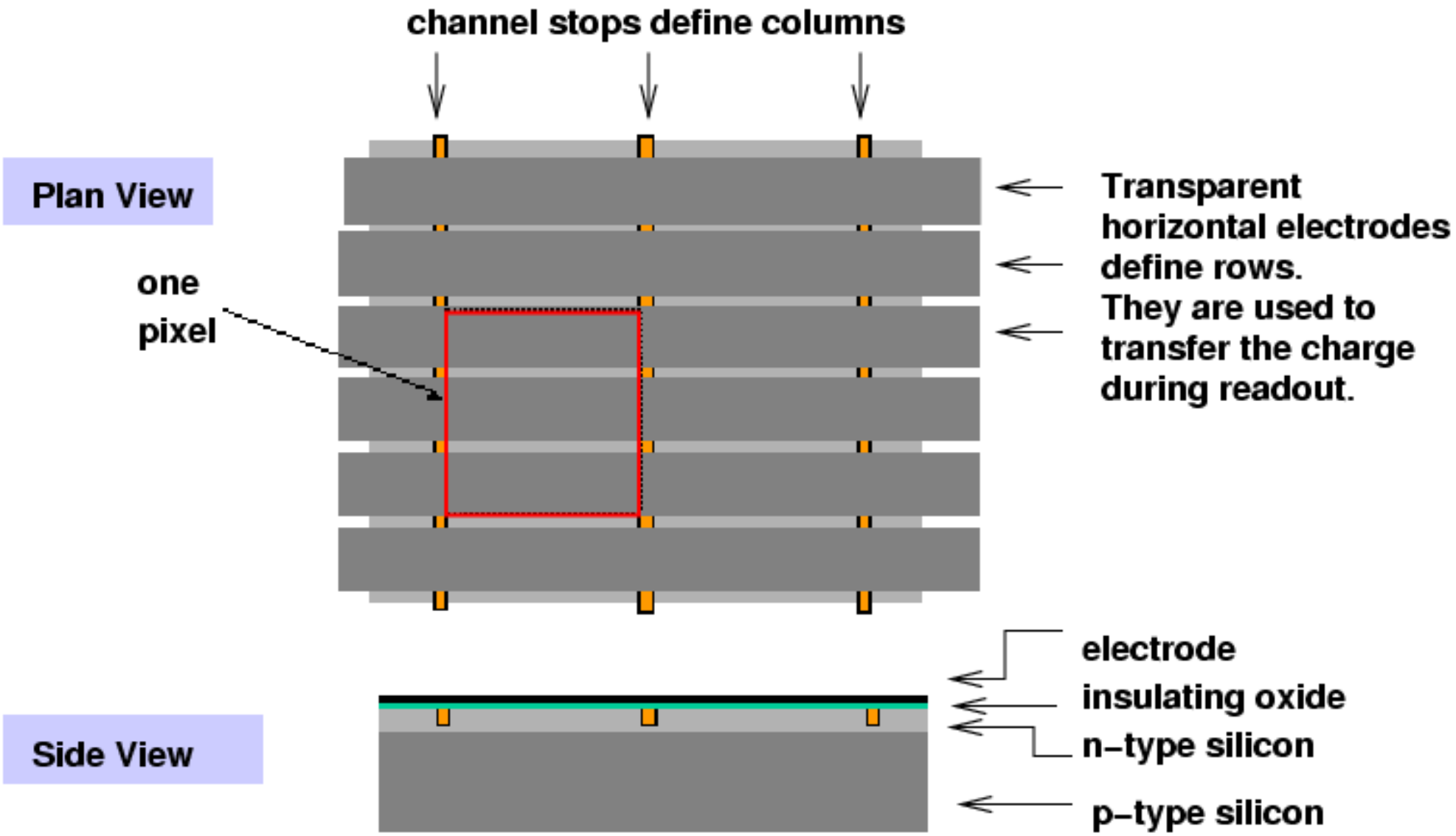
p-type silicon

n-type silicon

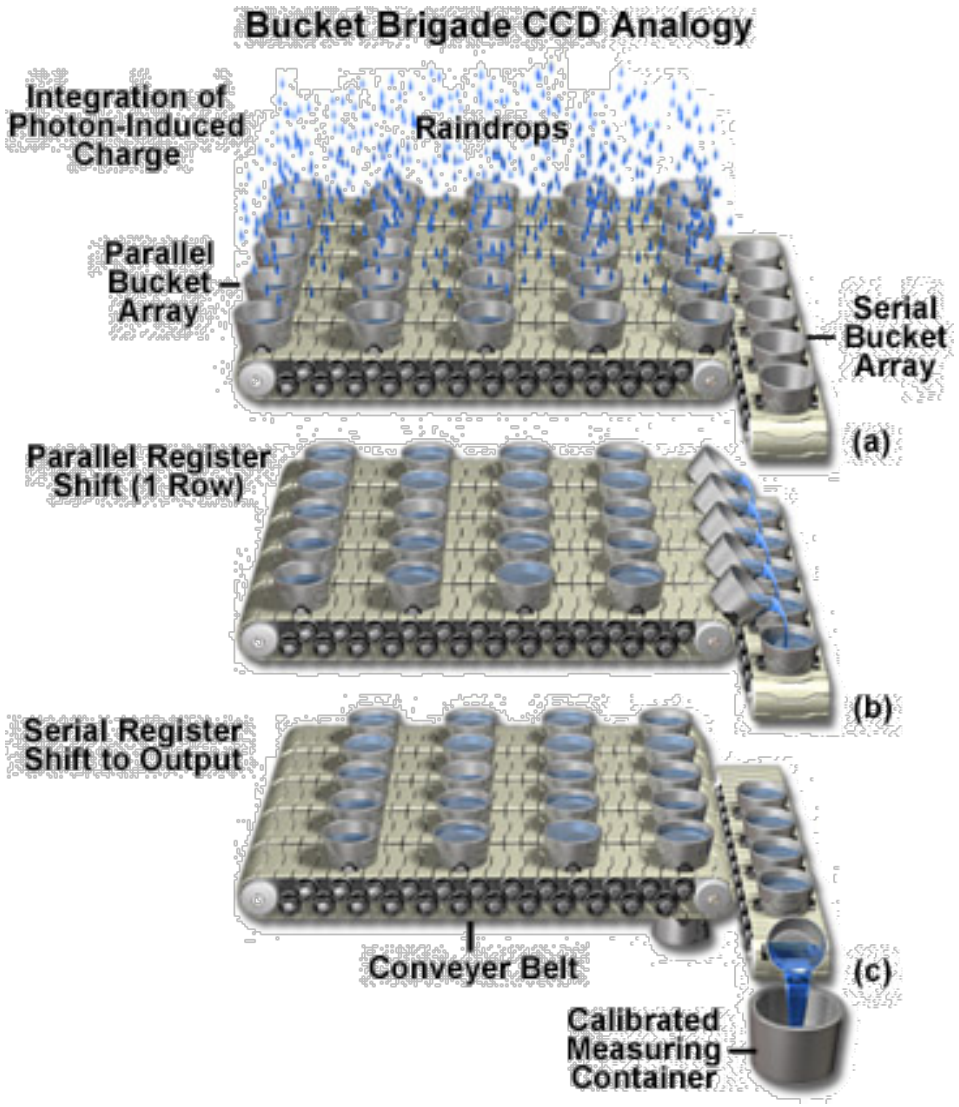
Silicon dioxide insulating layer

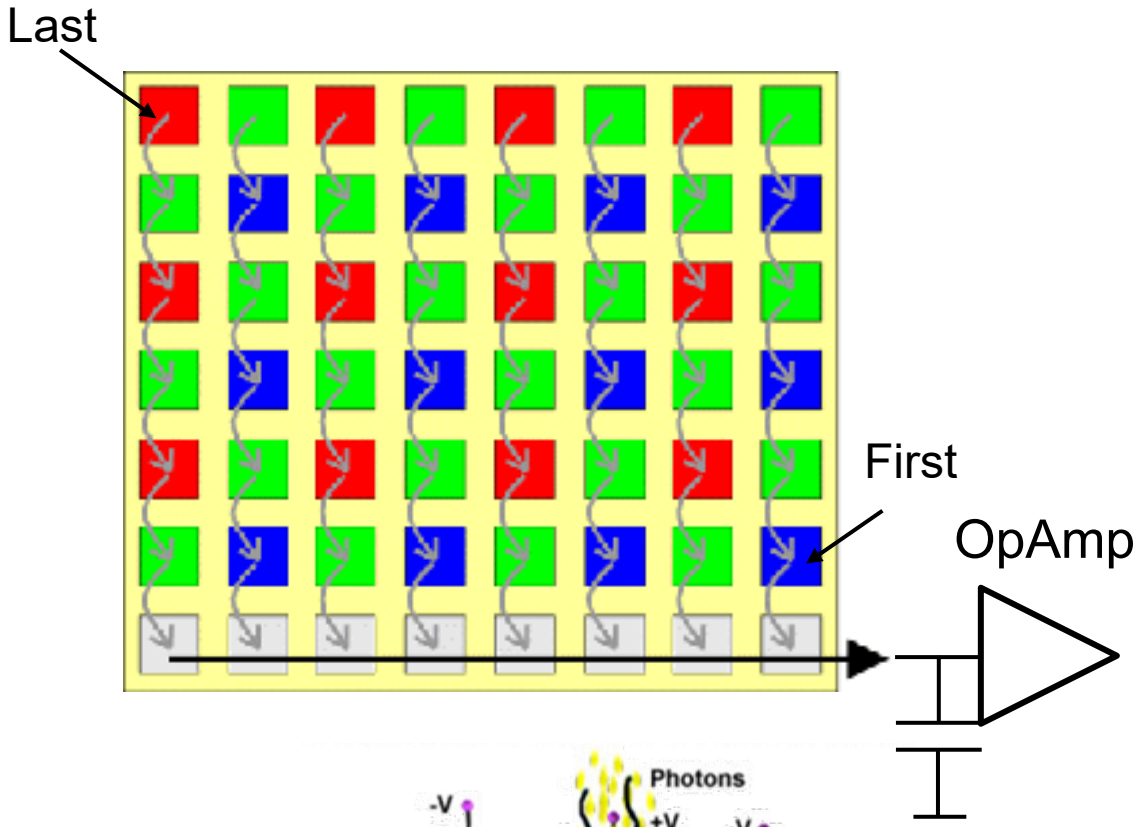
Polysilicon electrodes

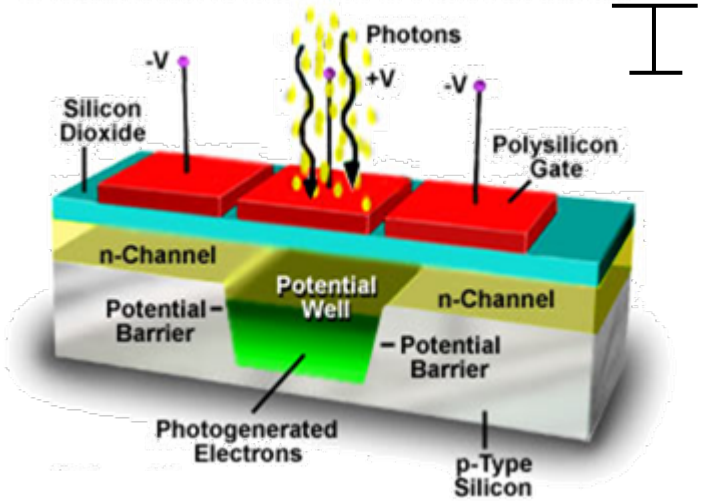
- 3D vertical integration
- anti-reflective coating


channel stops define columns
Plan View
one
pixel
Transparent
horizontal electrodes
define rows.
They are used to
transfer the charge
during readout.
Side View
electrode
insulating oxide
n–type silicon
p–type silicon

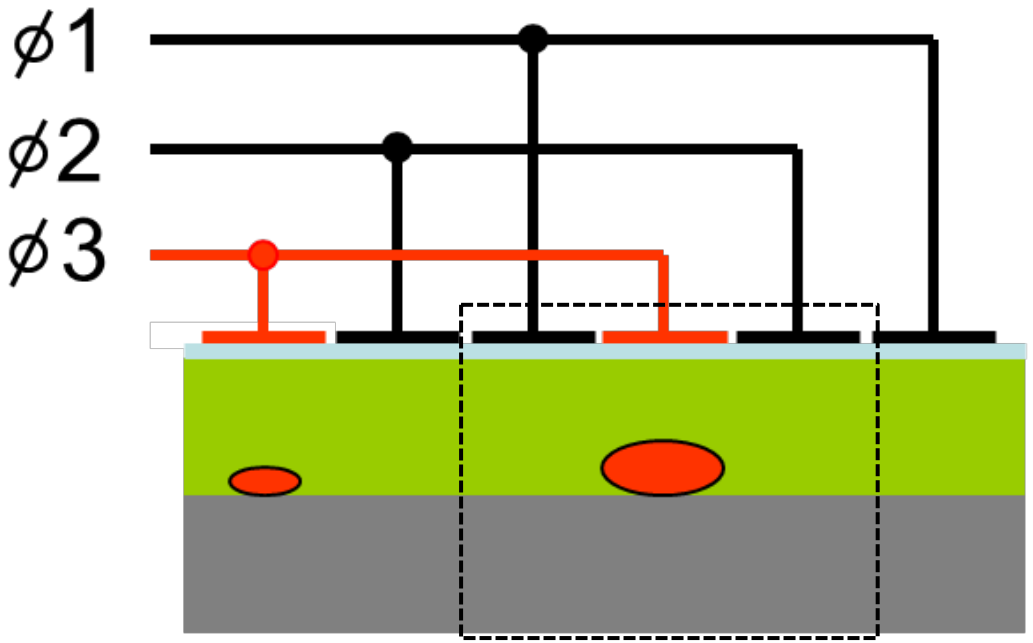
CCD readout

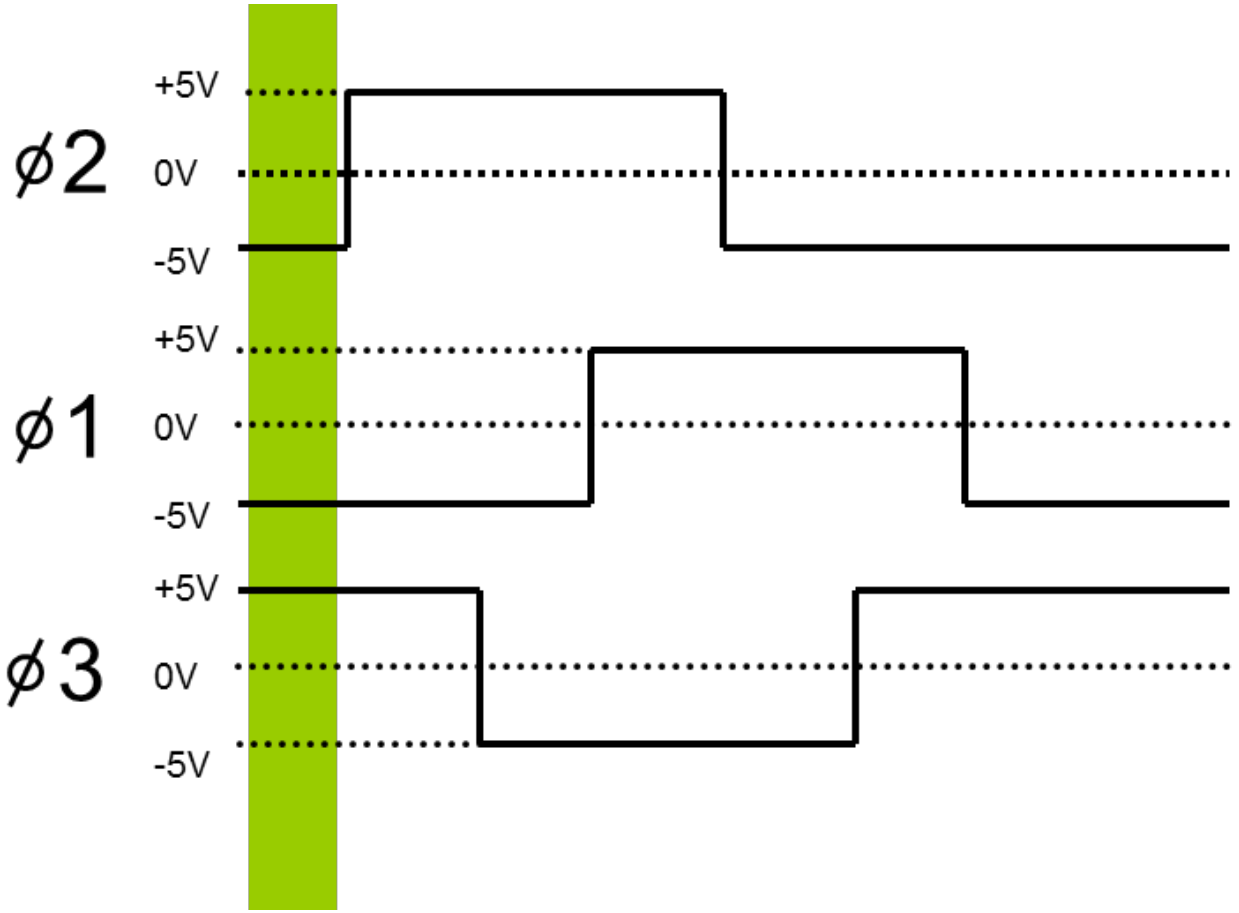

Bucket Brigade CCD Analogy
Integration of Photon-Induced Charge
Raindrops
Parallel Bucket Array
Serial Bucket Array
(a)
Parallel Register Shift (1 Row)
(b)
Serial Register Shift to Output
Conveyer Belt
(c)
Calibrated Measuring Container


Last
First
OpAmp

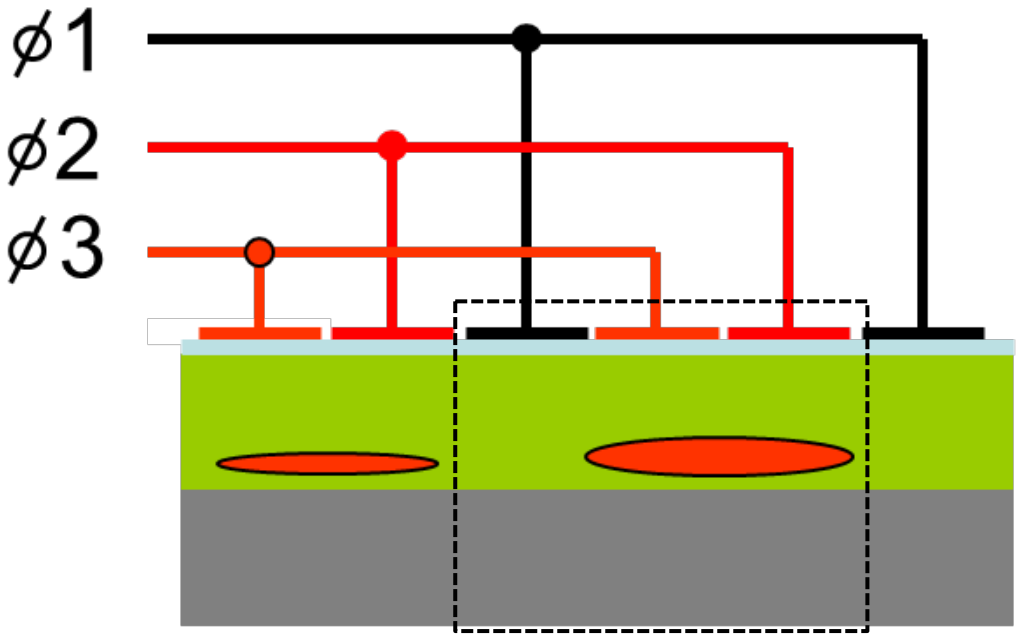

Photons
-V
+V
-V
Silicon Dioxide
Polysilicon Gate
n-Channel
Potential Barrier
Potential Well
n-Channel
Potential Barrier
Photogenerated Electrons
p-Type Silicon

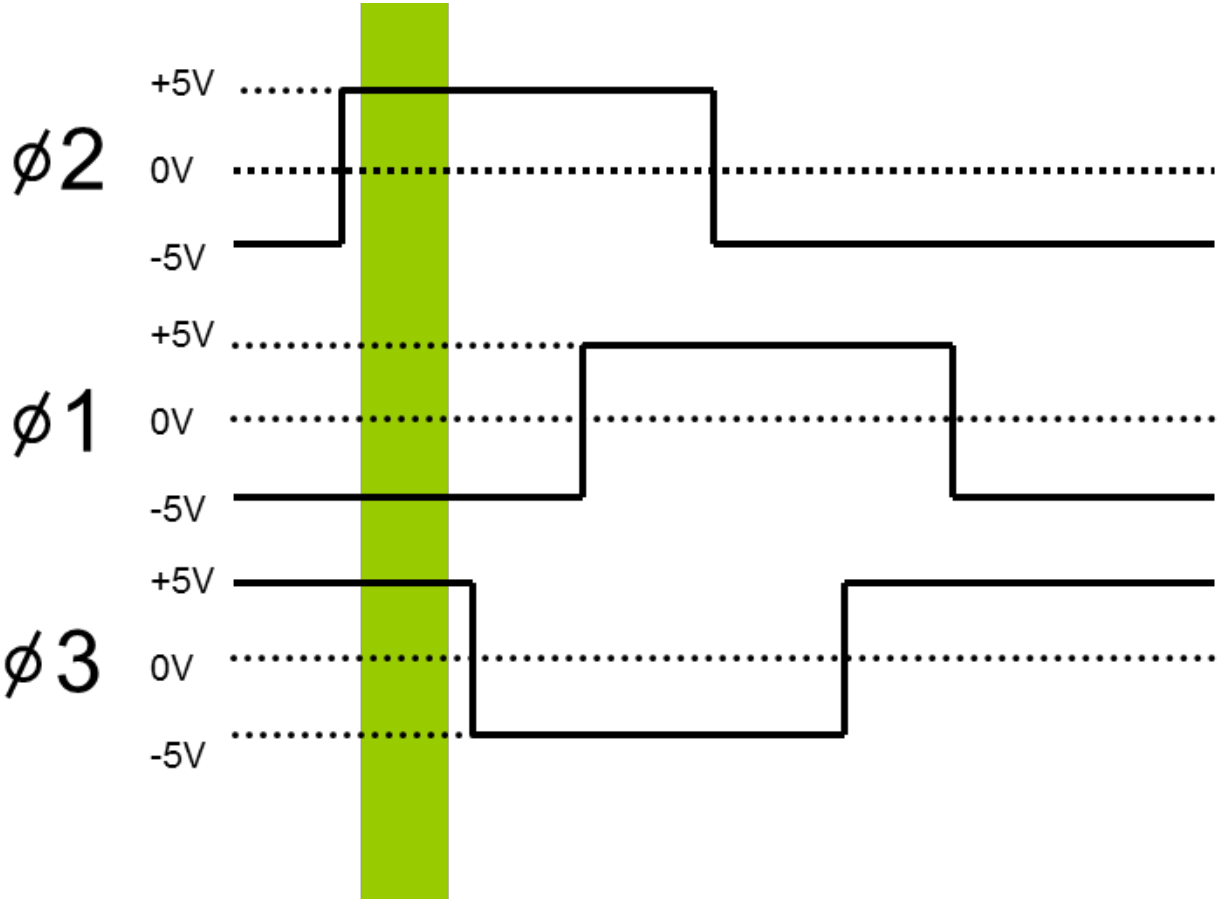
Readout – step 1



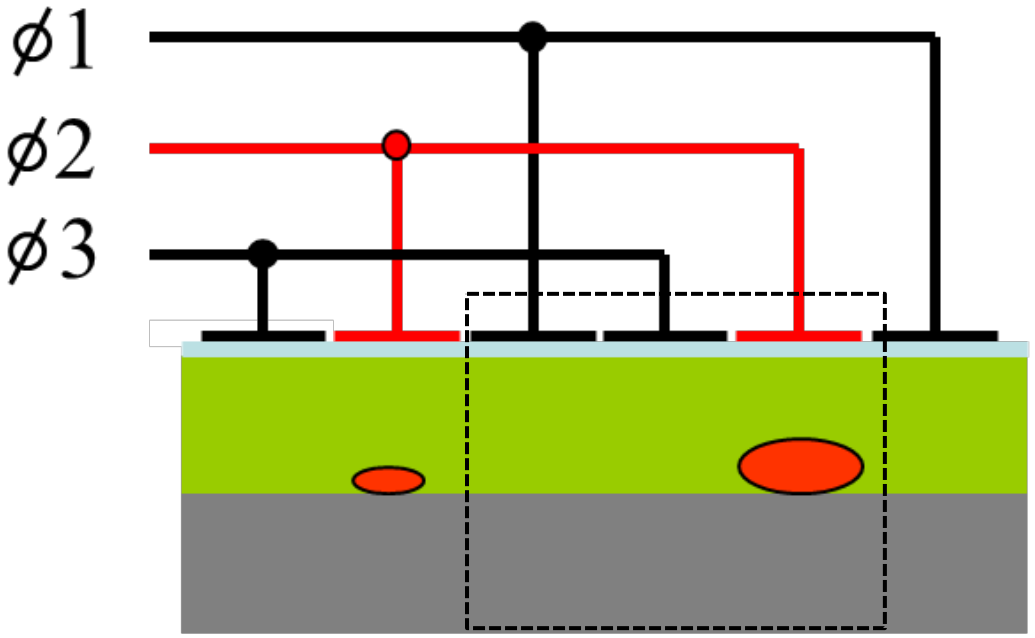


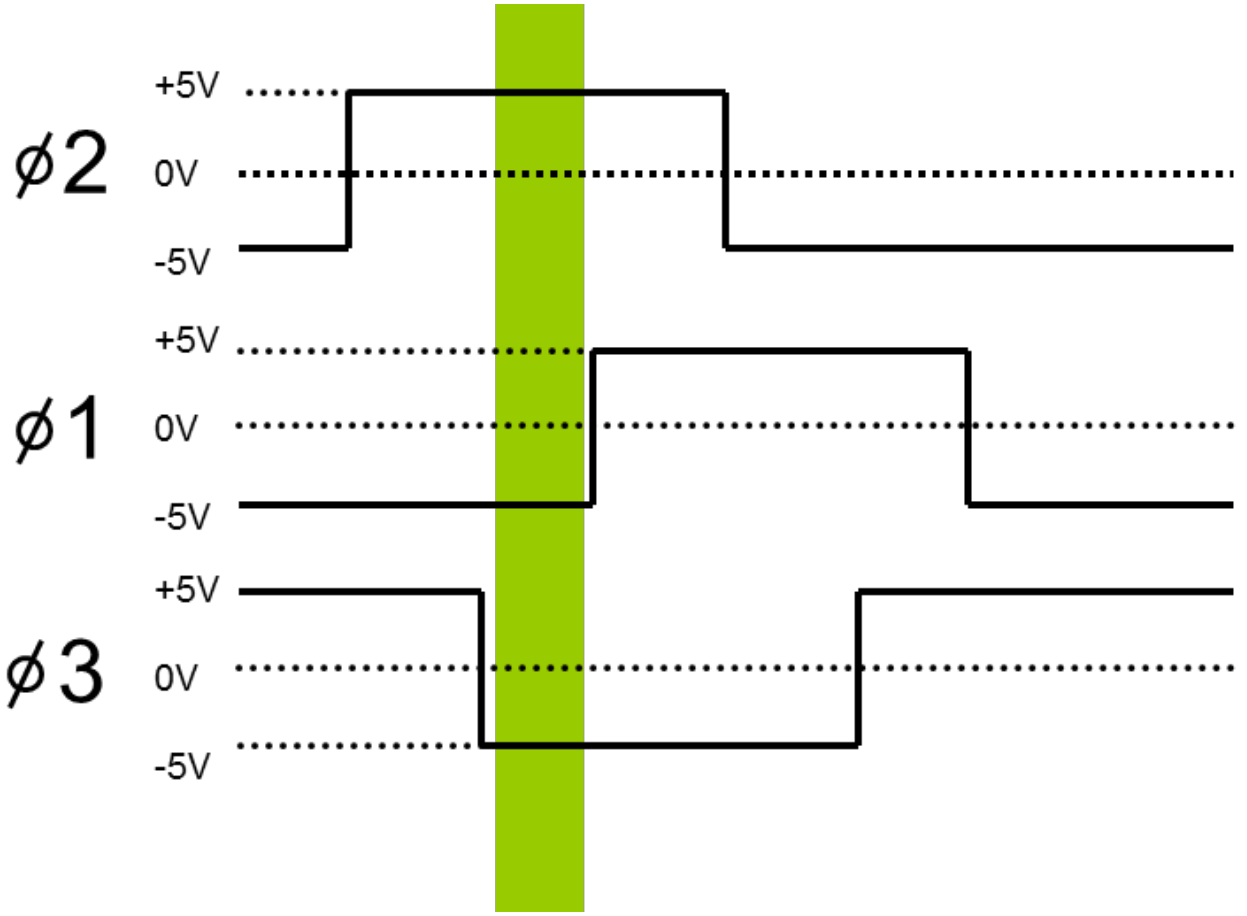
Readout – step 2



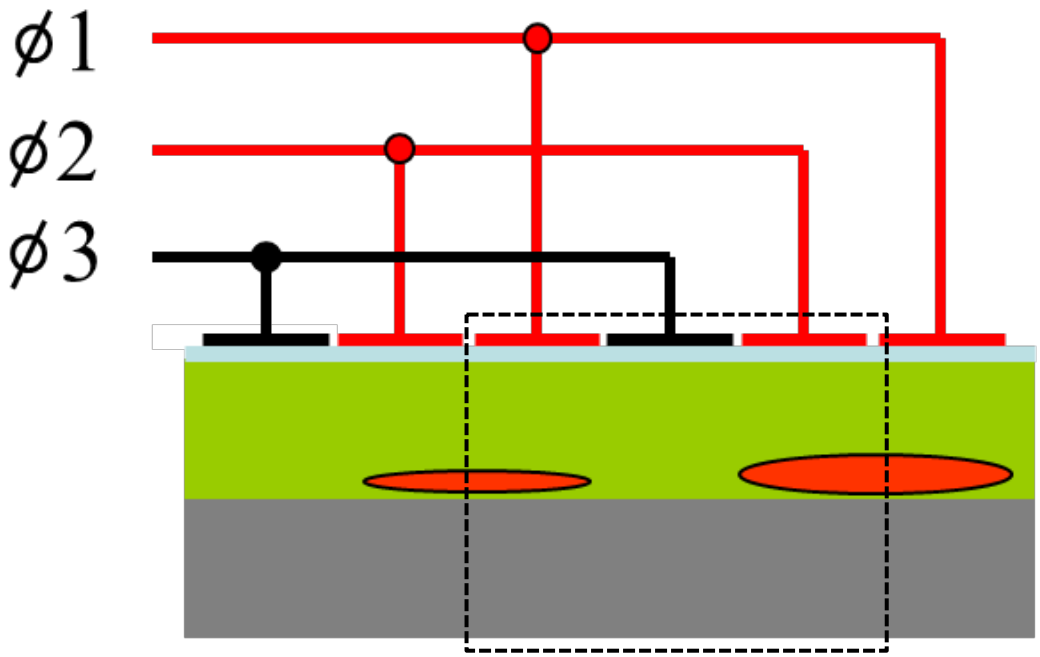


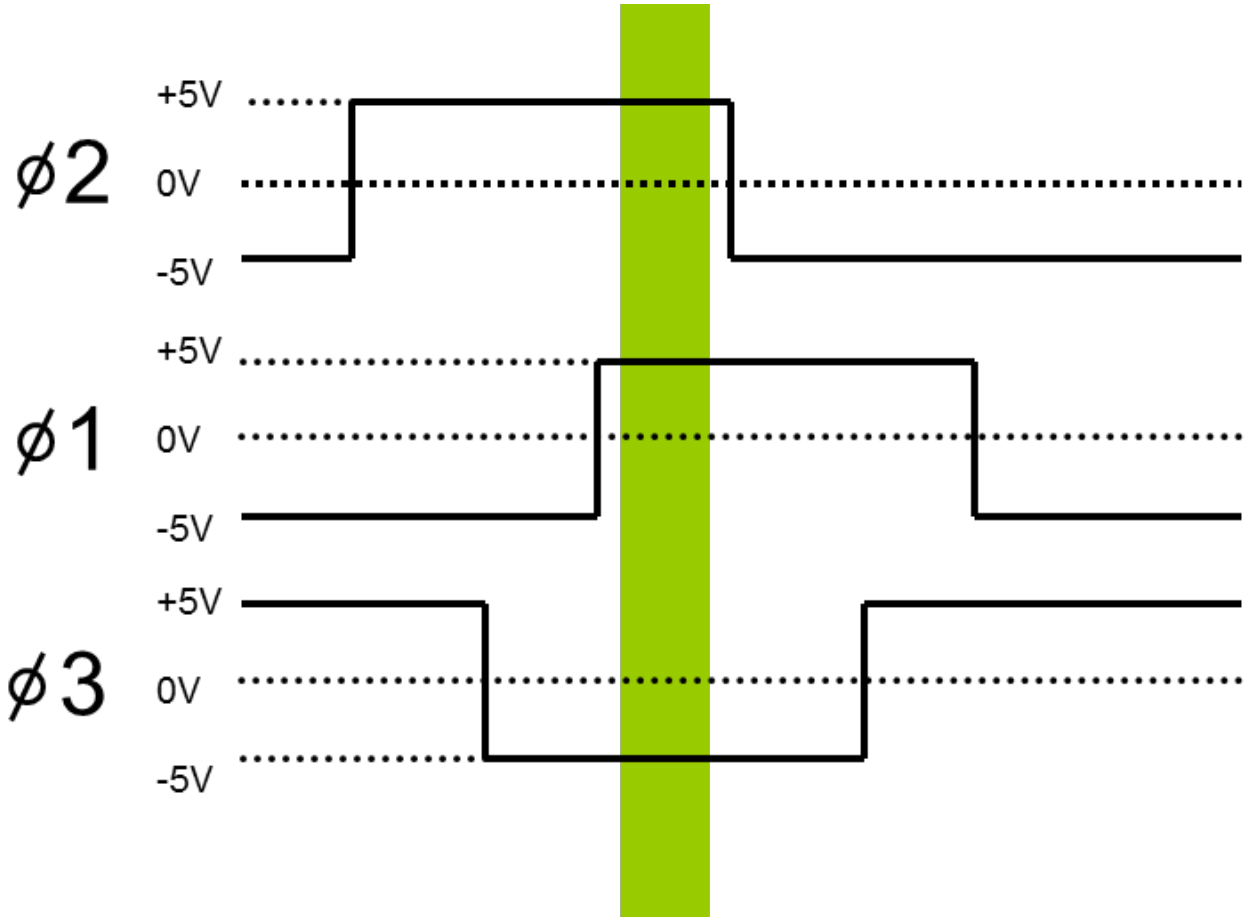
Readout – step 3



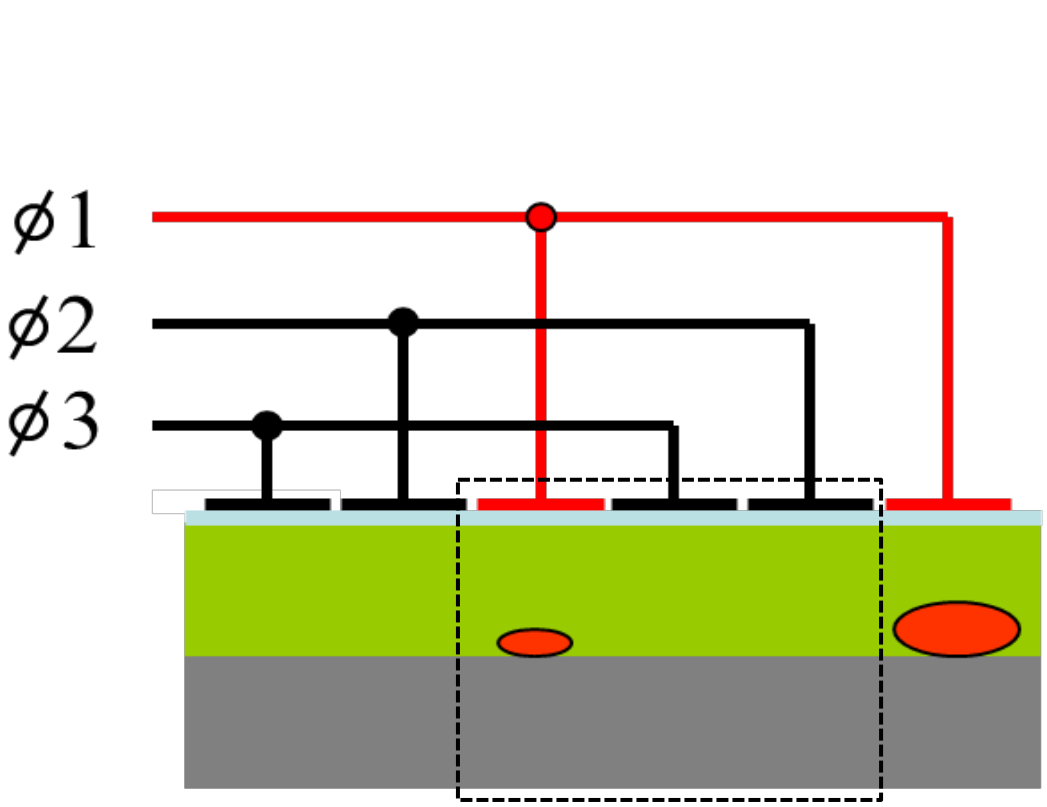


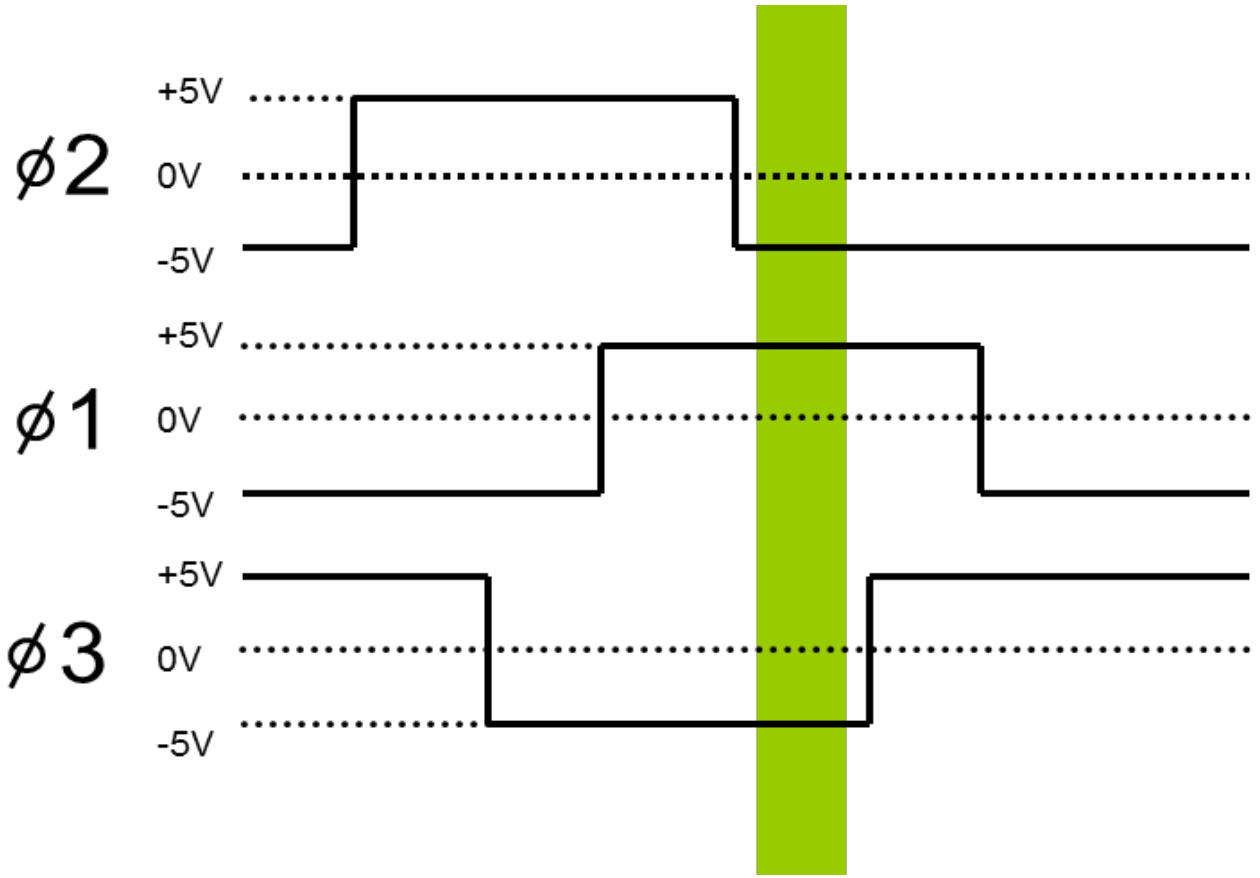
Readout – step 4



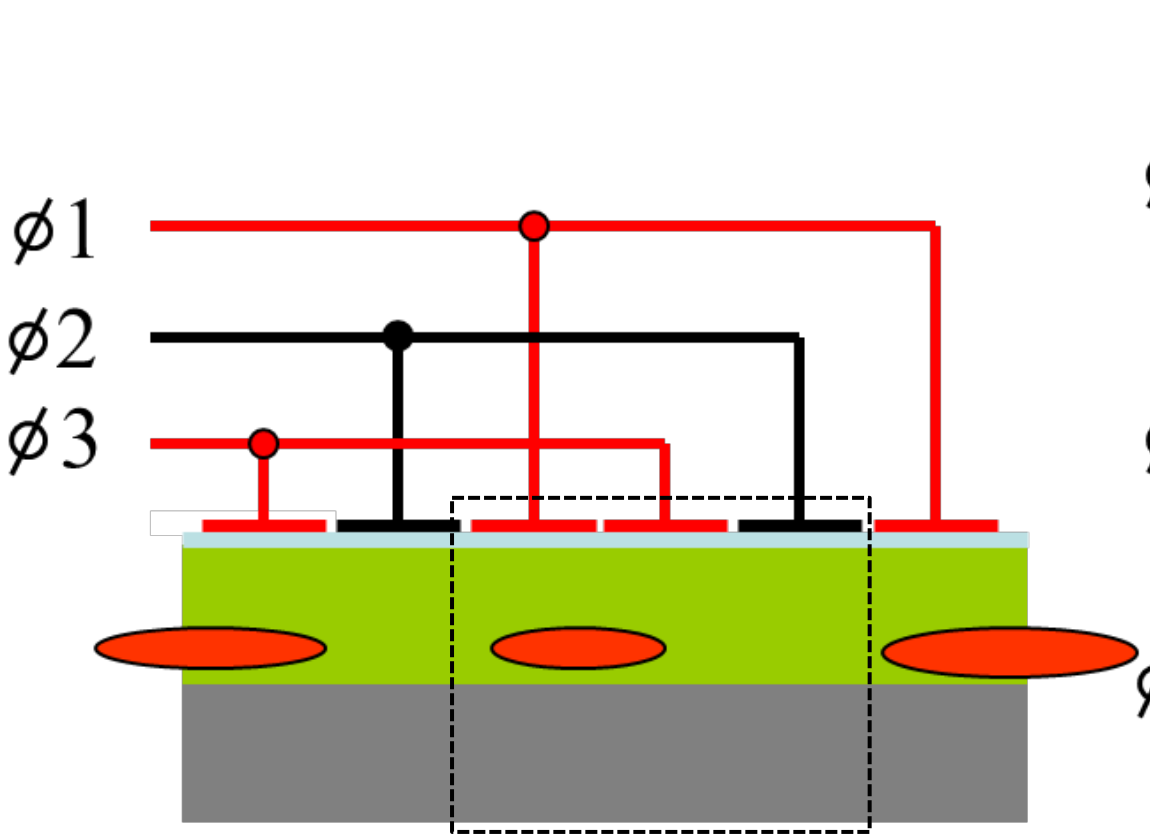


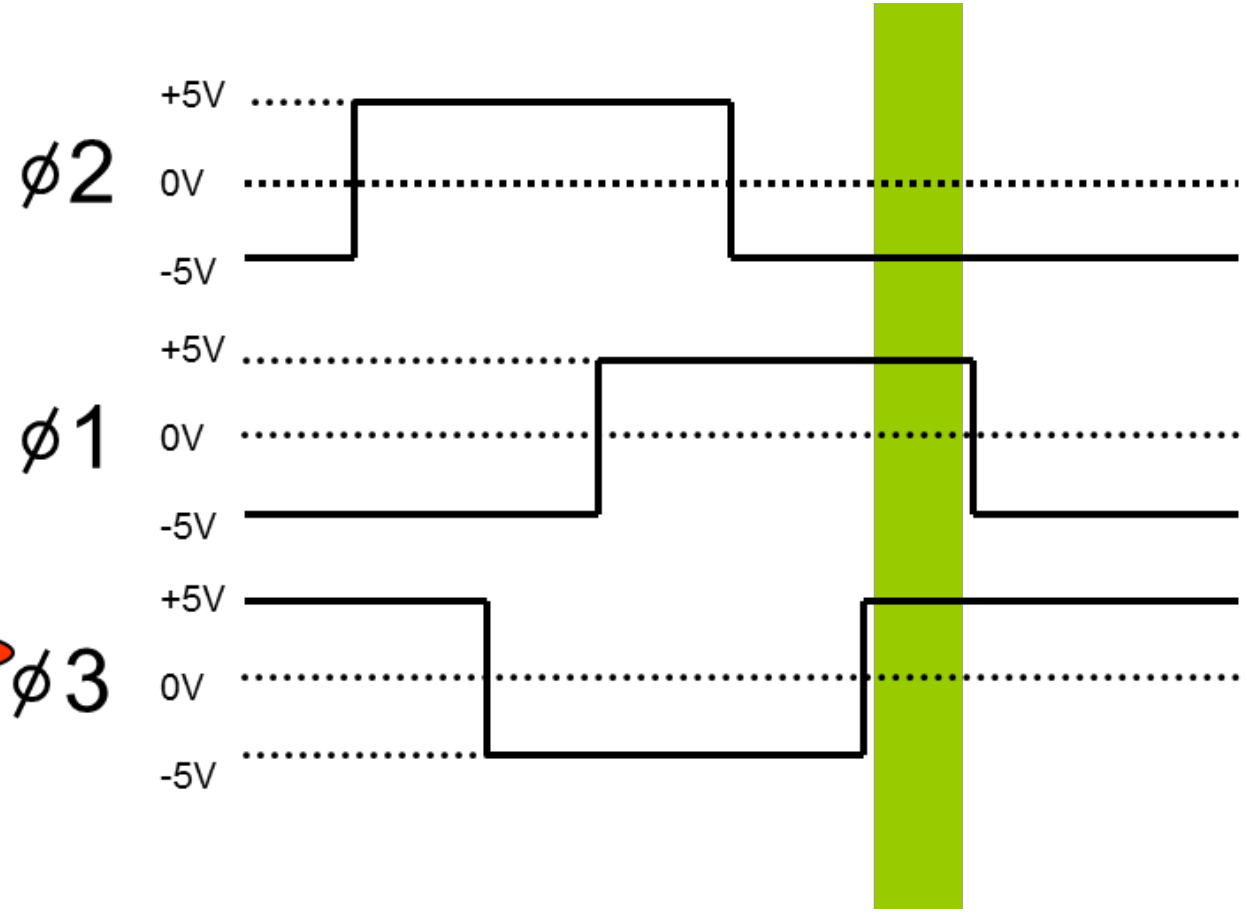
Readout – step 5





Readout – step 6





Charge generation

Quantum Efficiency (QE), Pixel Uniformity, Dark Current

Charge collection

Well Capacity, Pixels Size, Linearity, Blooming

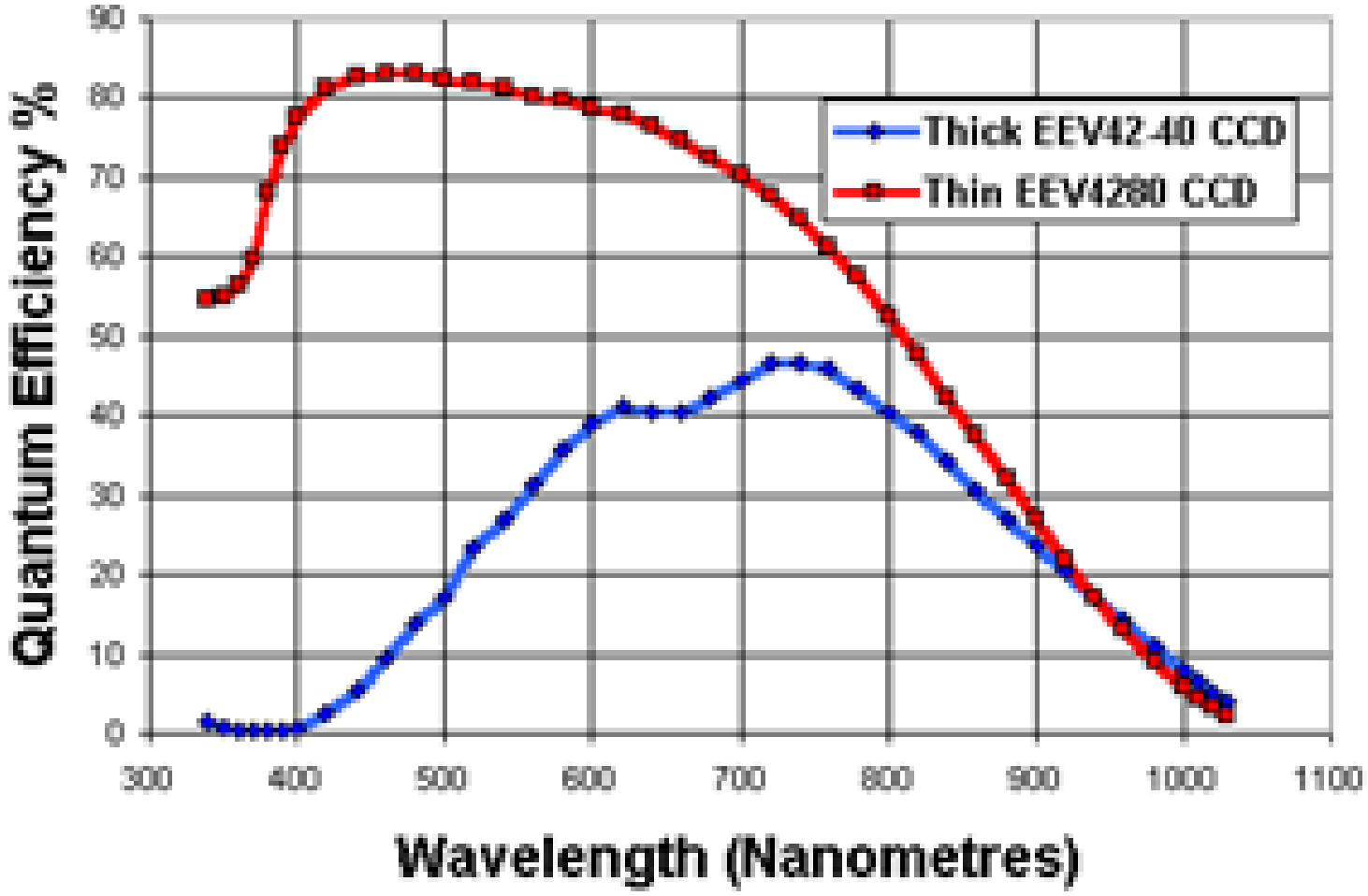
Charge transfer

Charge Transfer Efficiency (CTE), Defects

Charge readout

Readout Noise

Charge generation: Quantum Efficiency



$$QE = \frac{detected\ photons}{incident\ photons}$$

Probability that
a photoelectron will be released
for each incident photon

Mechanisms which hamper photon collection:

- Reflection
- Absorption
- Transmission

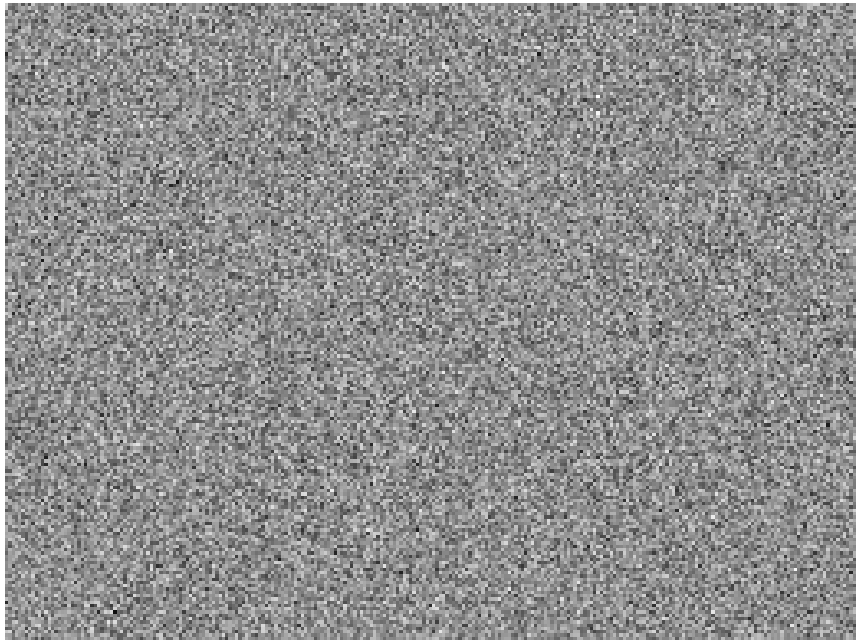
Charge generation: pixel uniformity

Fixed pattern noise

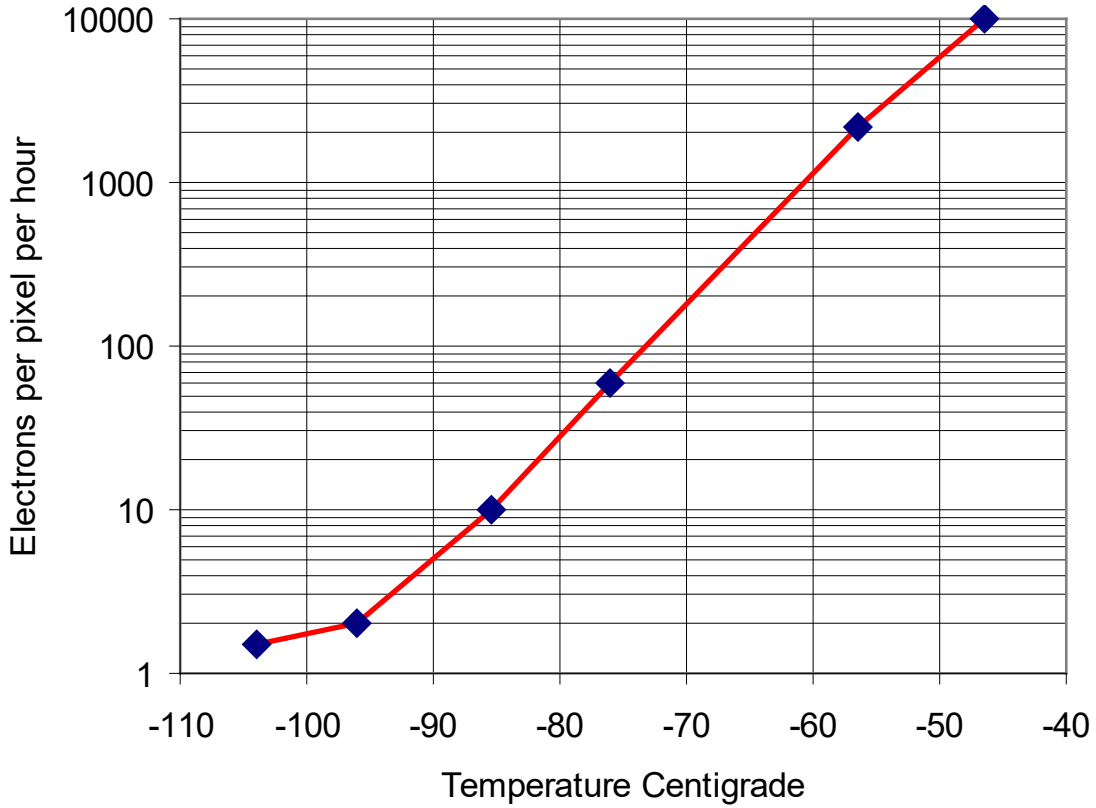
The sensitivity of pixels is not the same, for reasons such as differences in thickness, area of electrodes, doping, bias.

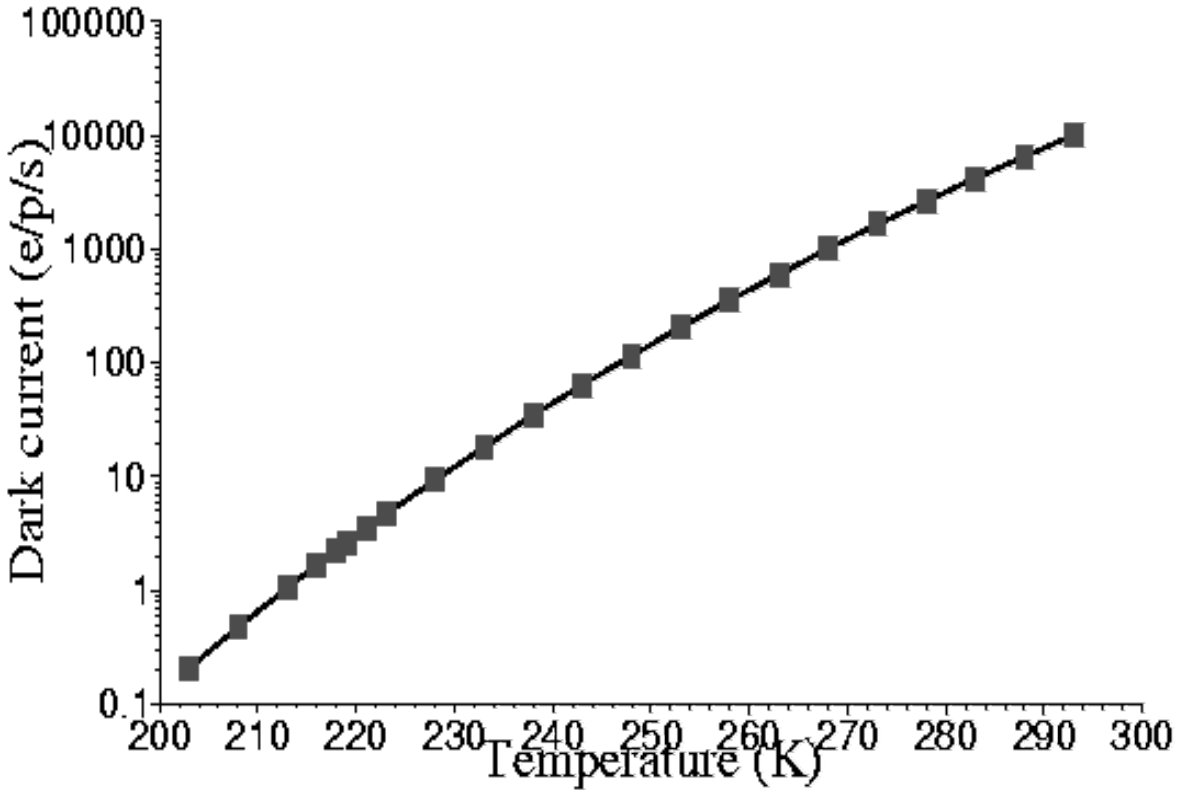
↓

However these differences do not change, and can be calibrated out by dividing by a flat field, which is an exposure of a uniform light source.



Charge generation: Dark Current





Thermal effects cause an electron to move from the valence band to the conduction band.

The majority of dark current is created near the interface between the Si and the SiO_2, where interface states at energy between the valence and conduction bands act as a stepping stone for electrons.

Well capacity = maximum charge that can be held in a pixel.

Typically, it is about 10,000 electrons/µm² → a few hundred thousand electrons per pixel

Example: V_{cell} = 5V, pixel size = 5µm× 5µm, C_{OX}/A = 35nF/cm²

$$\frac{Q}{A} = \frac{C_{ox}}{A} \cdot V_{cell} = 35\text{nF}/cm^2 \cdot 5V \approx 175nC/cm^2 \approx 11{,}000\ {}^{e}/_{\mu m^2}$$

$$\rightarrow Q \approx 300{,}000e$$

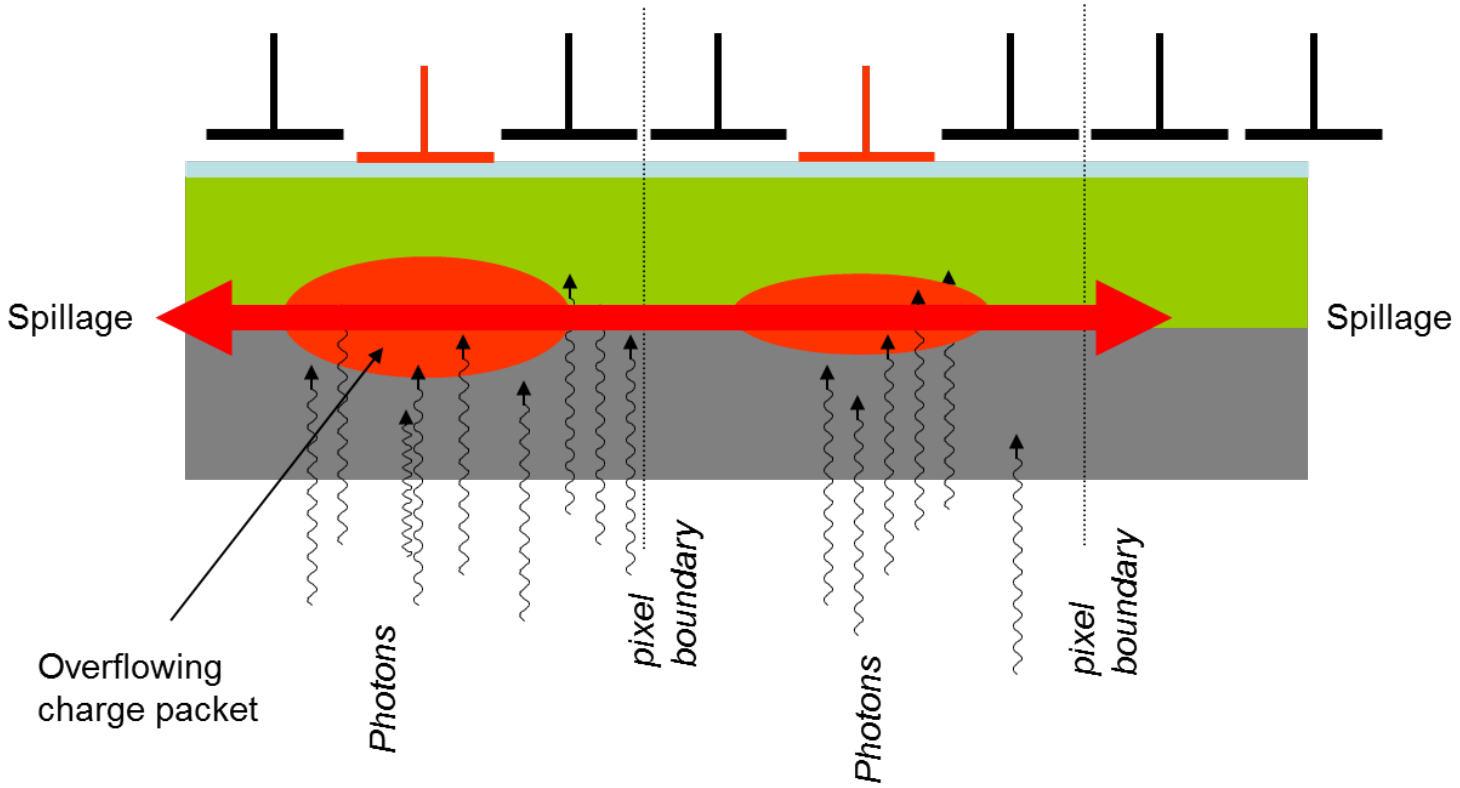
⇩

Saturation = when a pixel has accumulated the maximum amount of charge that it can hold.

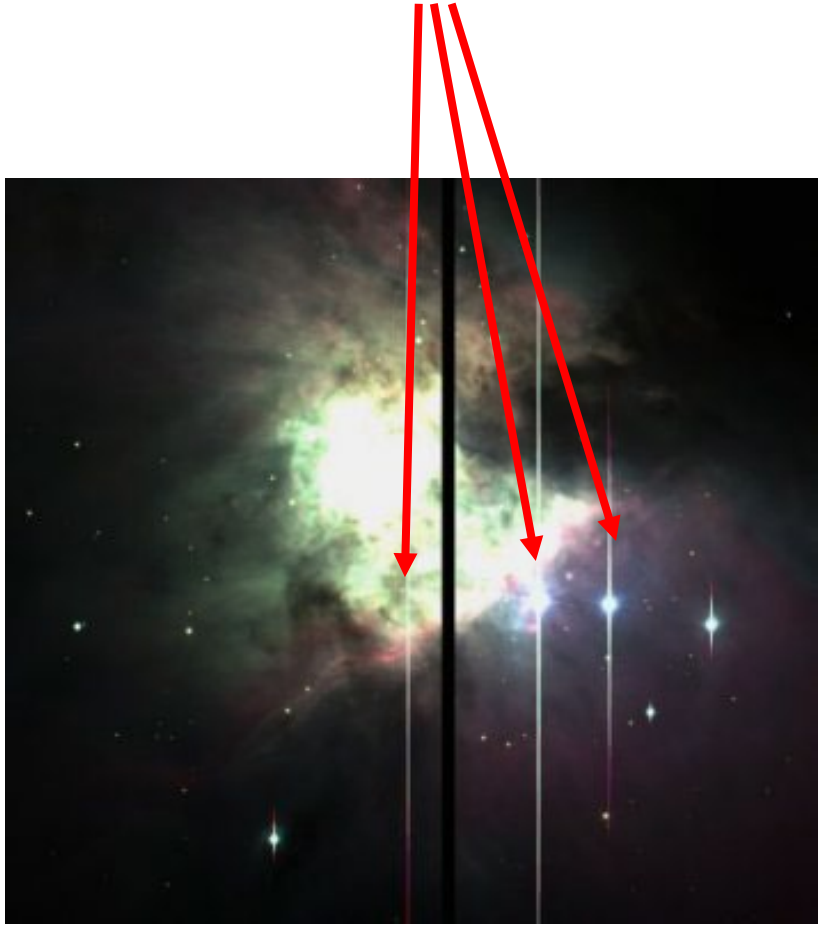
> 80% → non-linear response

> 100% → blooming

Charge collection: Blooming



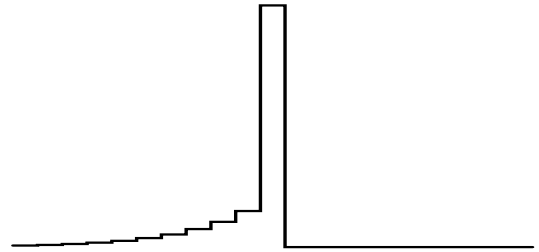
Bloomed star images



Blooming occurs mainly vertically, as there is little horizontal bleeding because of the permanent doped channel stops.

Charge Transfer Efficiency (CTE)

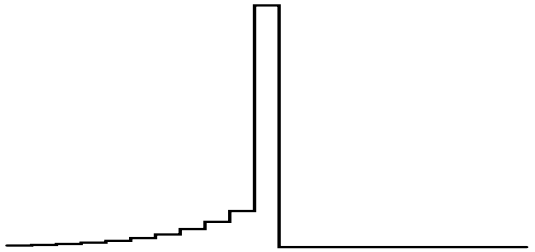
= fraction of electrons transferred from one pixel to the next
(typically 0.9999 to 0.999999)



Charge Transfer Inefficiency (CTI)

= fraction of electrons deferred by one pixel
CTI = 1 – CTE (typically 10^{-6} to 10^{-4})

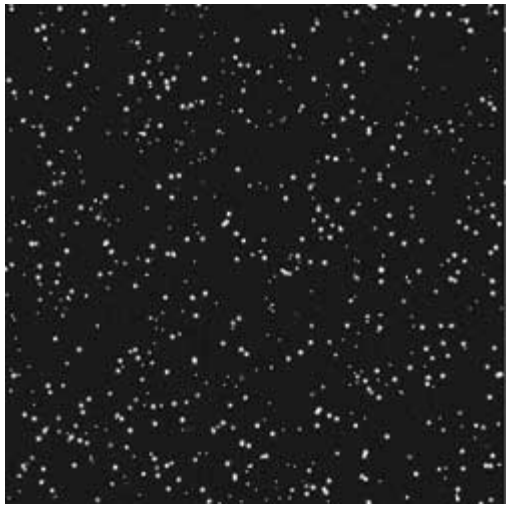
→ charges are trapped by defects in the silicon crystal lattice



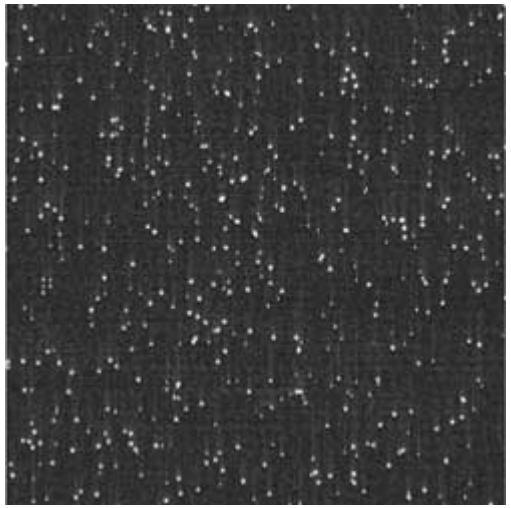
Example: CTE = 0.99999, 12Mpixel (about 3.5 kpixel × 3.5 kpixel)

CTI in the worst case (about 7,000 transfers):

$$CTI_{TOT} = 1 - \left(0.99999^{7,000}\right) = 6.7\%$$

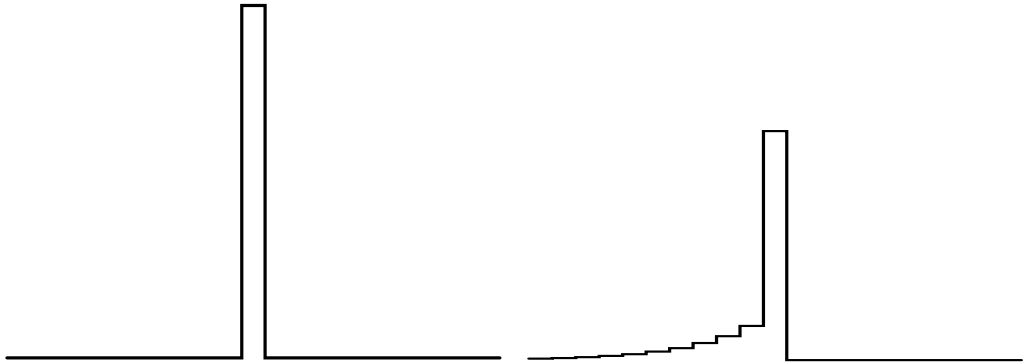


good CTE

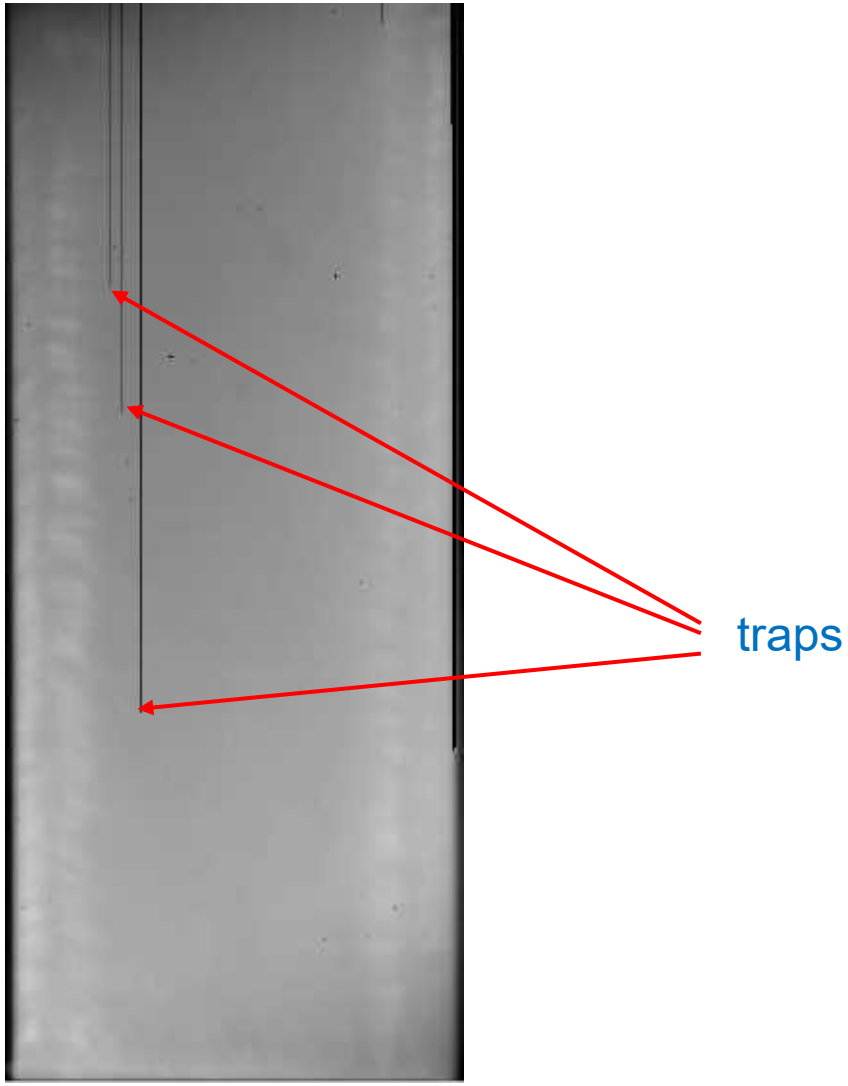


bad CTE

Part of the original image is smeared with an exponential decay function, producing “tails”:



original image after *n* transfers

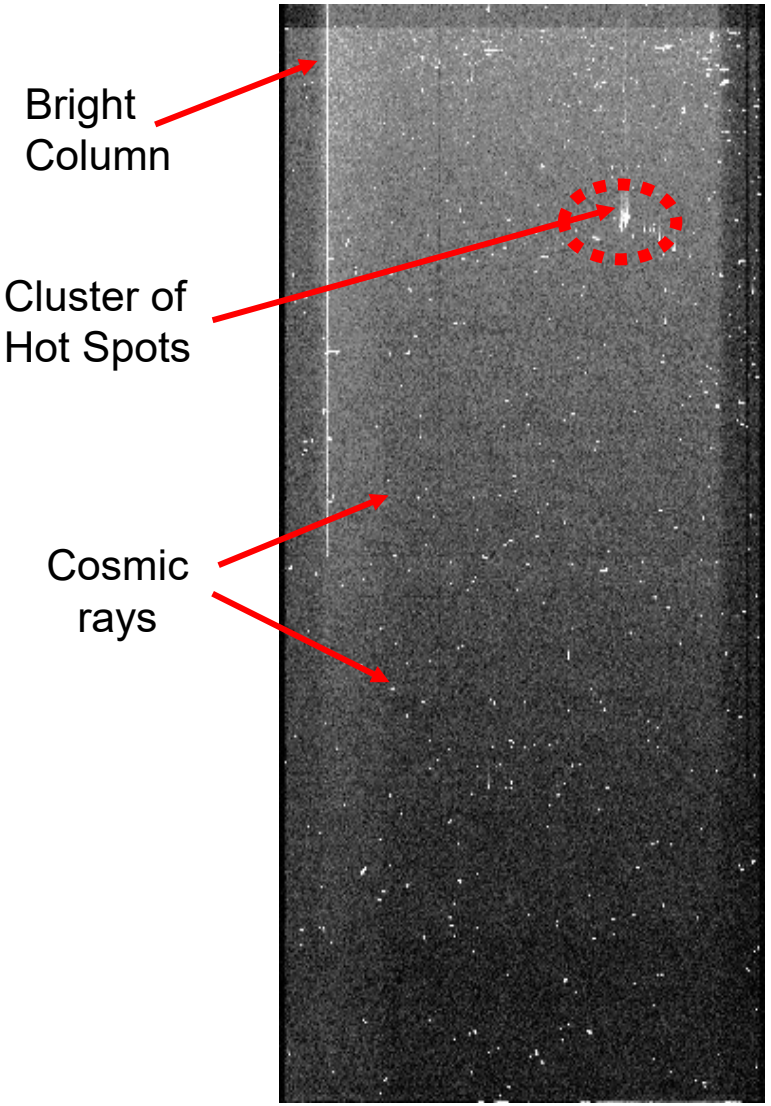


Dark columns

Defects caused by “traps” that block the vertical **transfer** of charge during image readout.

Dark columns are removed by calibration.

Other defects



Bright columns

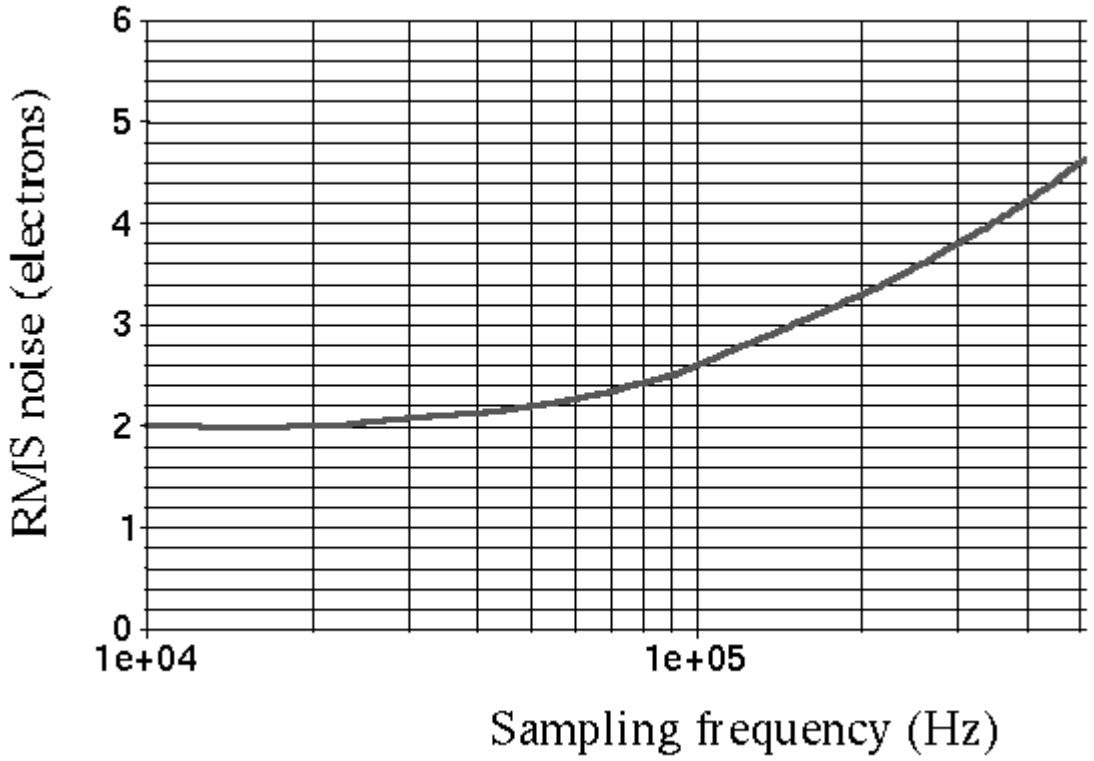
Caused by traps. Electrons contained in such traps can leak out during **transfer** causing a vertical streak.

Hot Spots

Pixels with higher than normal **dark current**. Their brightness increases linearly with exposure times.

Somewhat rarer are light-emitting defects which are hot spots that act as tiny LEDs and cause a halo of light on the chip.

Readout noise (1/3)



Mainly due to amplifier noise.

It can be reduced by reducing the bandwidth

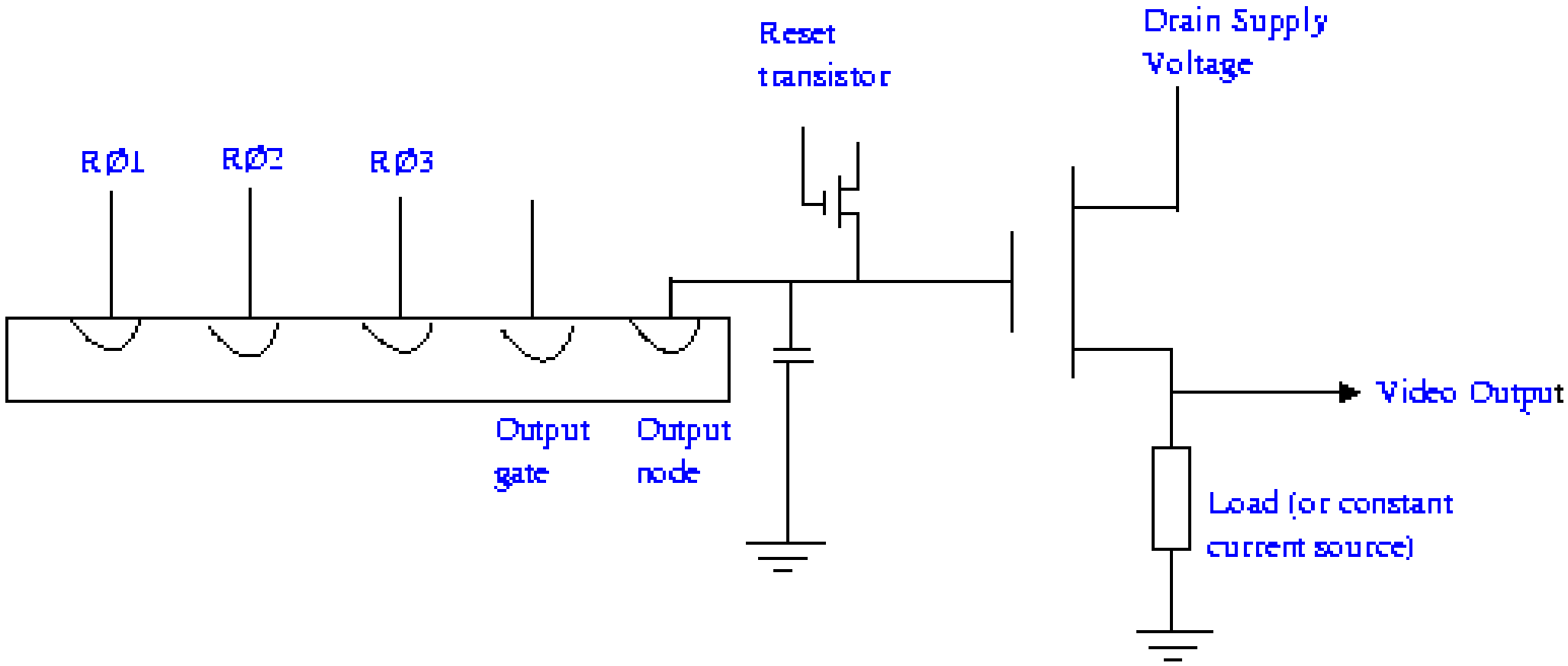
(slower readout).

= rate at which one pixel is read by the CCD

f_s = 500kHz → T_s = 2µs

Assuming 12Mpx → $T_{readout}$ = 2µs × 12Mpx = 24s !!

The readout circuit consists of a storage capacitor, a reset transistor and a voltage amplifier or buffer (in this example a source follower).



Reset noise

Noise associated with recharging the output storage capacitor:

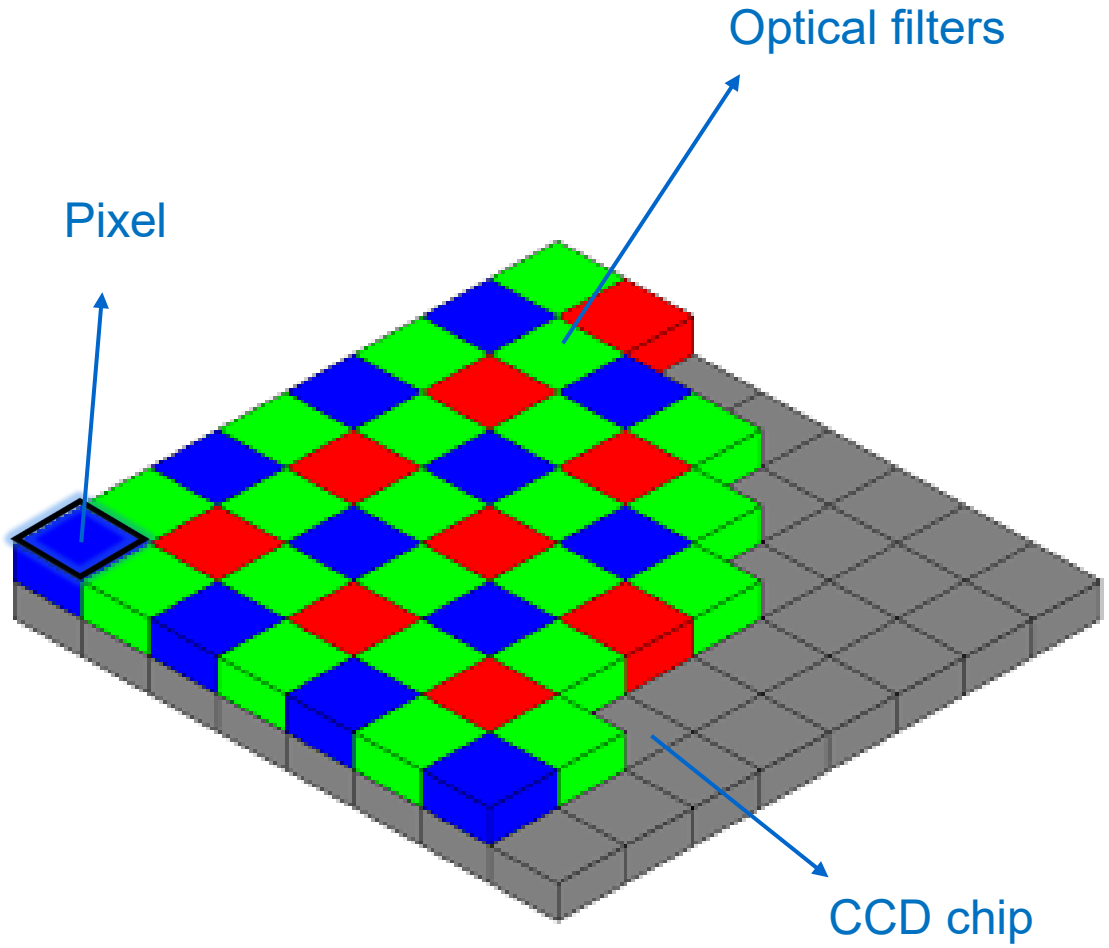
$$\sigma Q_{reset} = \sqrt{kTC}$$

where C is the output capacitance.

⇩

This is removed by correlated double sampling

→ The output voltage is measured after reset and again after readout. The first value is subtracted from the second.



Bayer array

= most common RGB color filter

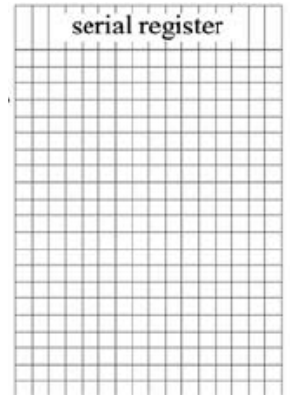
Since the human eye is more sensitive to green than to the other two colors, the Bayer array has twice as many green color filters.

This also means that with the Bayer array, the human eye can detect more detail than if the three colors were used in equal measures in the filter.

Demosaicing algorithm are used to “fill in” missing color information for each pixel

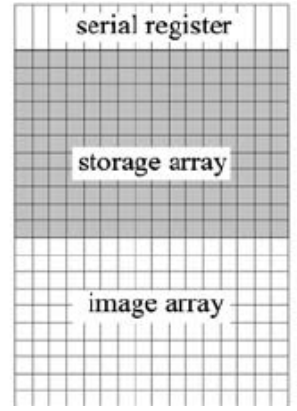
Full frame entire chip exposed to light

→ long readout time, signal corrupted during readout (shutter is needed)



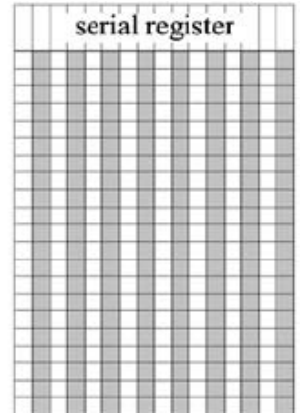
Frame transfer the chip upper (or lower) part is protected against light and the image is only made in the lower part of the chip.

Fast transfer from uncovered to masked region + slow readout from masked region.



Interline transfer every other line is protected from light.

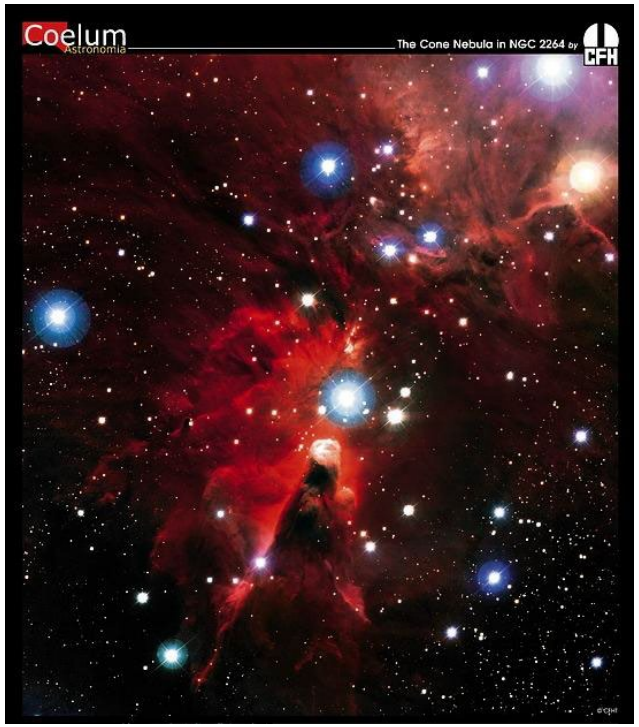
Uncovered line is transferred to masked line for readout.



CCD applications

- Digital cameras
- Optical scanners
- Microscopy and biology
- Astrophotography (galaxies, nebulae)
- Infrared photography (night vision, VLSI testing)











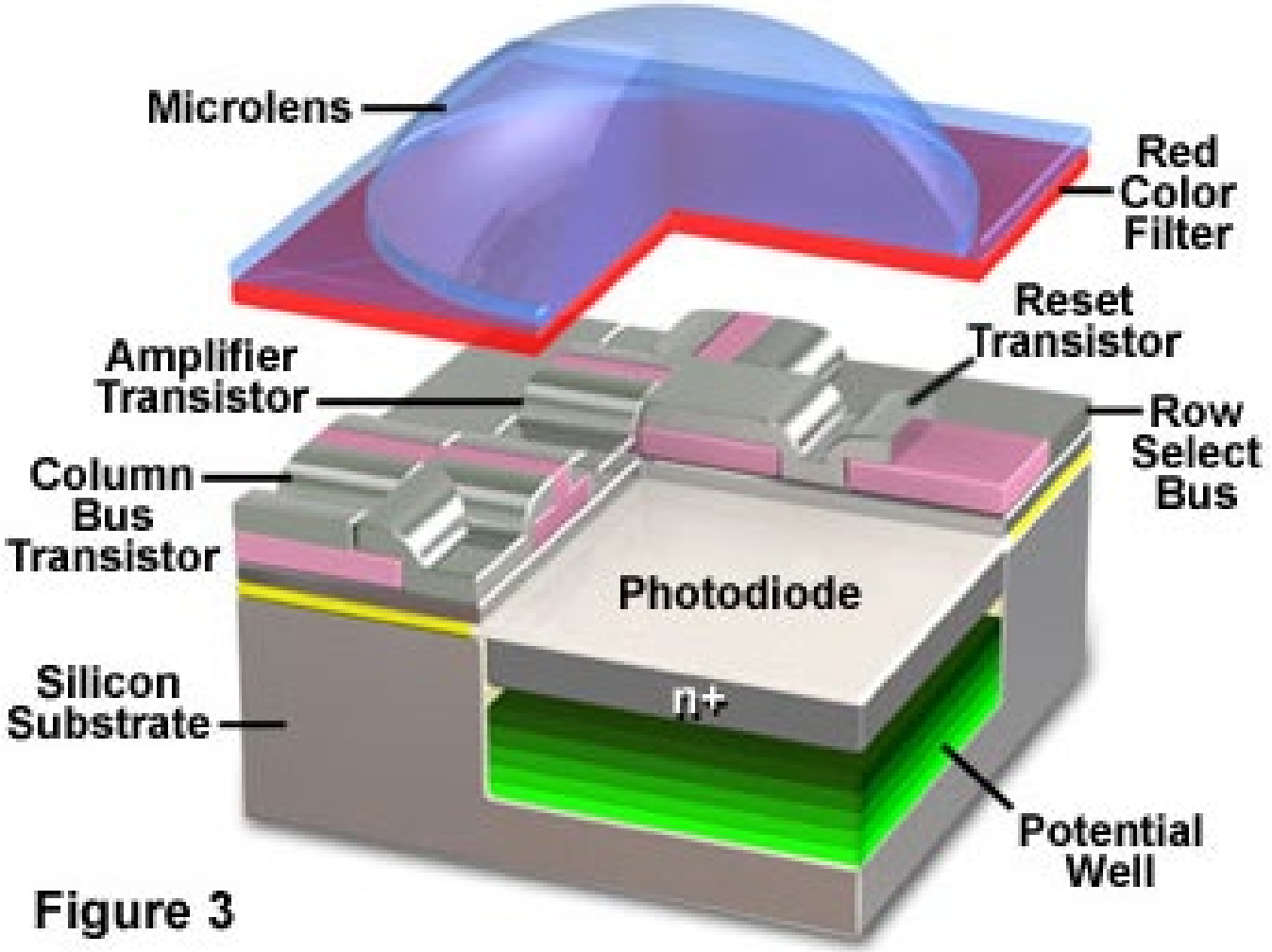
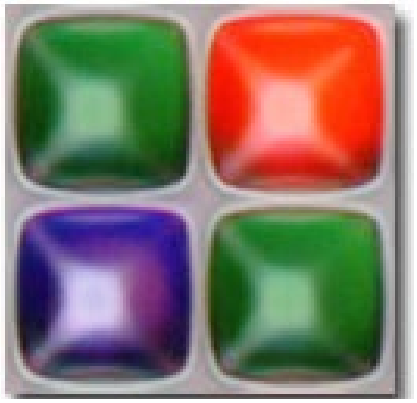
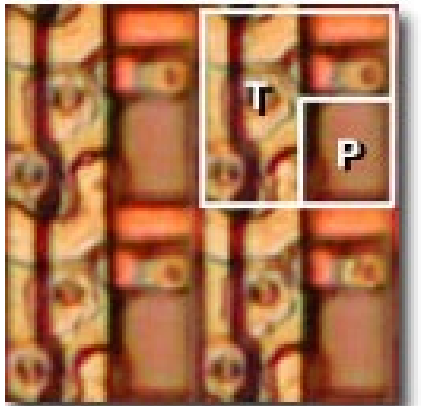

Microlens
Red
Color
Filter
Reset
Transistor
Amplifier
Transistor
Row
Select
Bus
Column
Bus
Transistor
Photodiode
n+
Silicon
Substrate
Potential
Well

Figure 3





T ≈ 70%

P ≈ 30%

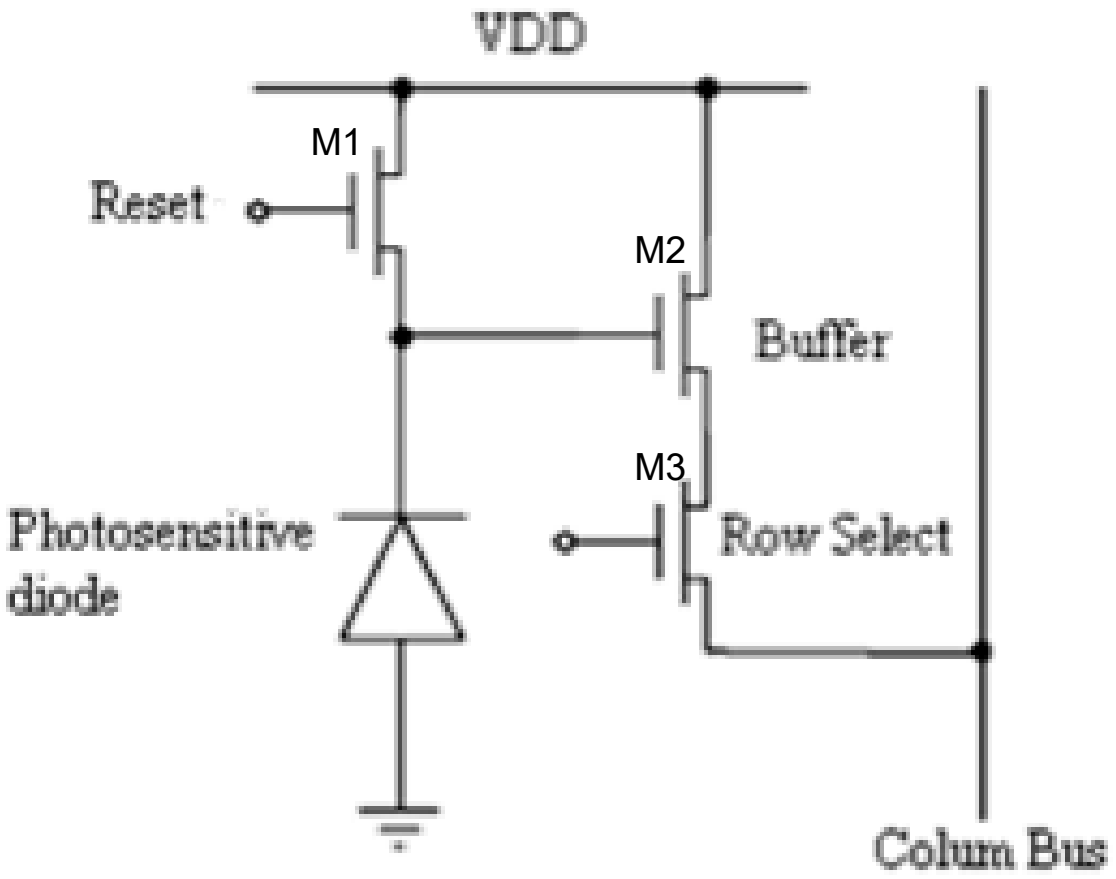
FF ≈ 30%

Both the **photodiode** and **readout amplifier** are incorporated into each pixel

This enables the charge accumulated by the photodiode to be converted into an amplified voltage inside the pixel and then transferred in sequential rows and columns to the analog signal-processing portion of the chip.

Main components of pixels:

- Photodiode
- 3 transistors
- Busses
- Bayer filter
- Micro-lens



M1

Reset Transistor

to restore initial condition before exposure.

M2

Amplifier transistor - source follower.

The load being common to all the pixels in a column.

M3

Row-select transistor

It connects the pixel output to the column bus for readout.

All the pixels in a particular column are connected to one column amplifier.

- Fixed Pattern Noise (FPN) caused by:
  - Variation in photodiode efficiency
  - Mismatches in source follower transistor performance

  FPN is constant and reproducible
  → flat field correction (= on chip subtraction of an image with constant illumination)

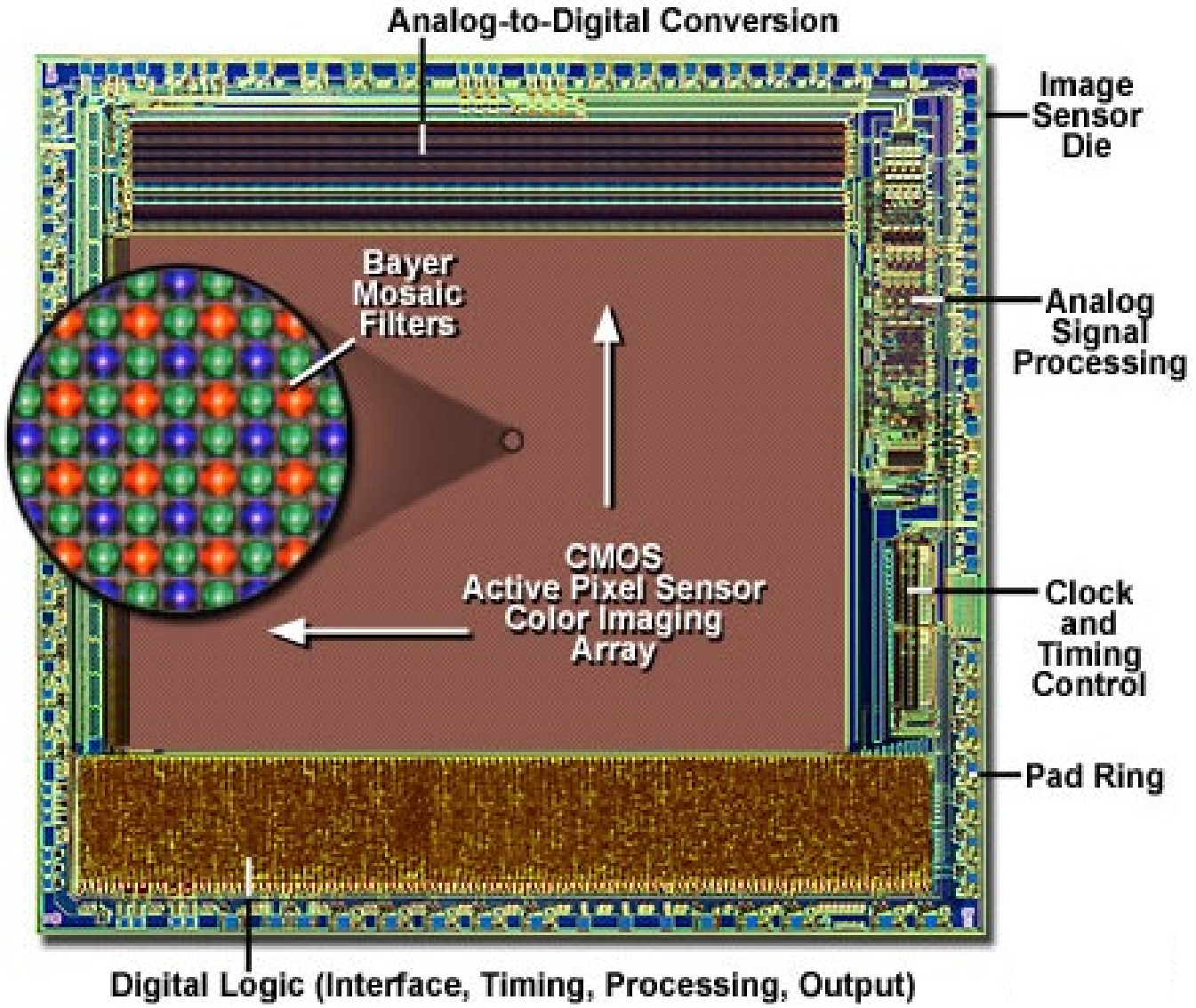
- Reset noise (kTC noise)

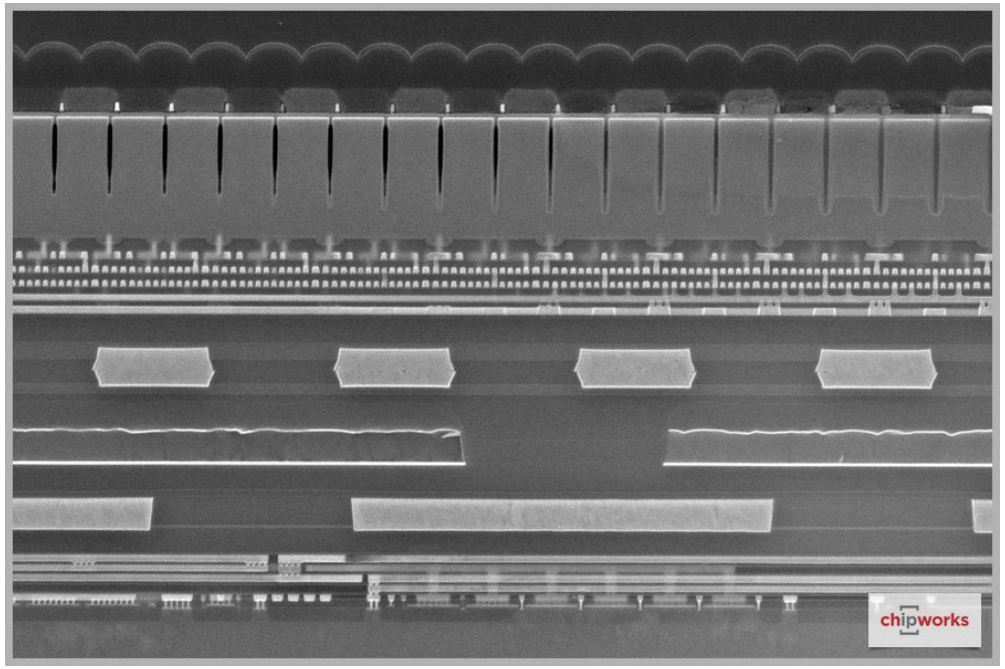
$$v_n = \sqrt{\frac{KT}{C}}$$

- Amplifier noise (1/f noise)

- Photodiode shot noise

CMOS sensor general structure



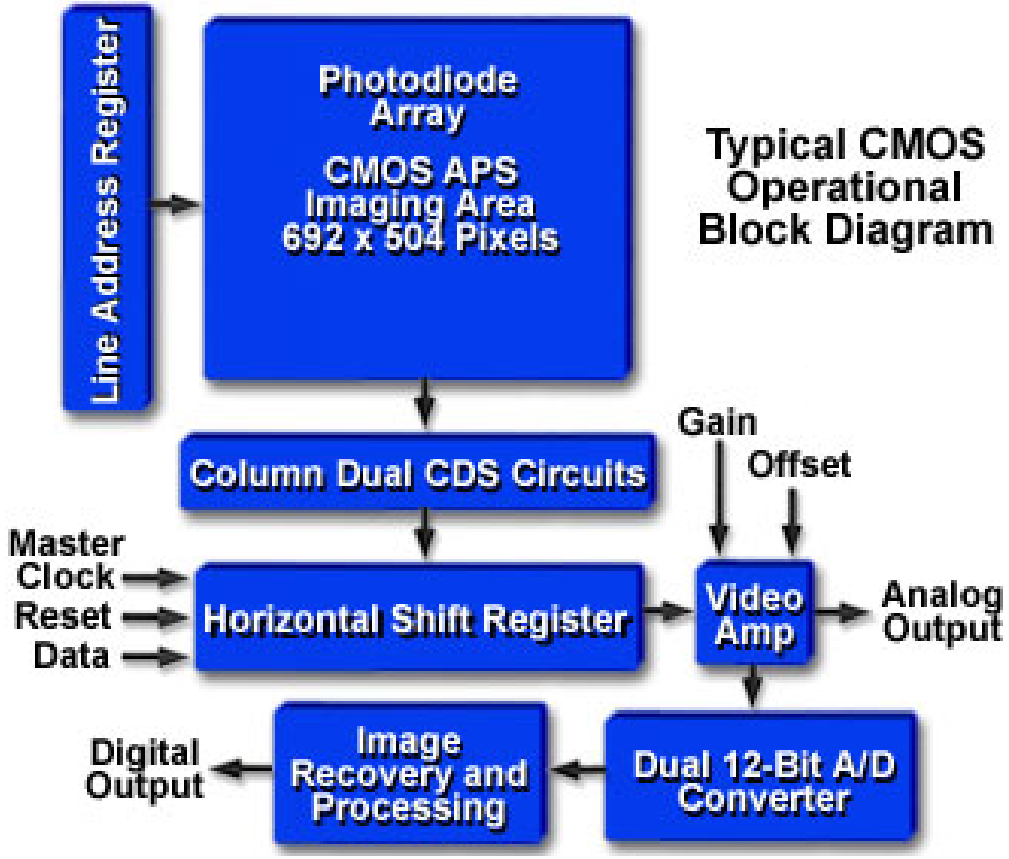


Back-illuminated CMOS image sensor implemented with a 3D-stacked hybrid-bonding (CIS + ISP) solution:

CIS = CMOS Image Sensor

ISP = Image Signal Processor

- VGA resolution (640x480)
  + shielded pixels for black level compensation
- Line address register for readout
- Correlated Double Sampling circuit
  to reduce the pixel reset noise
- Horizontal Shift Register
- Video Amplifier with adjustable gain and offset
- ADC converter
- Image processing
  (interpolation, smoothing, white and black balancing…)



Alternative:
One Video Amp + ADC for each column
(high frame rate)

The most commonly used digital interfaces for CMOS sensors readout are:

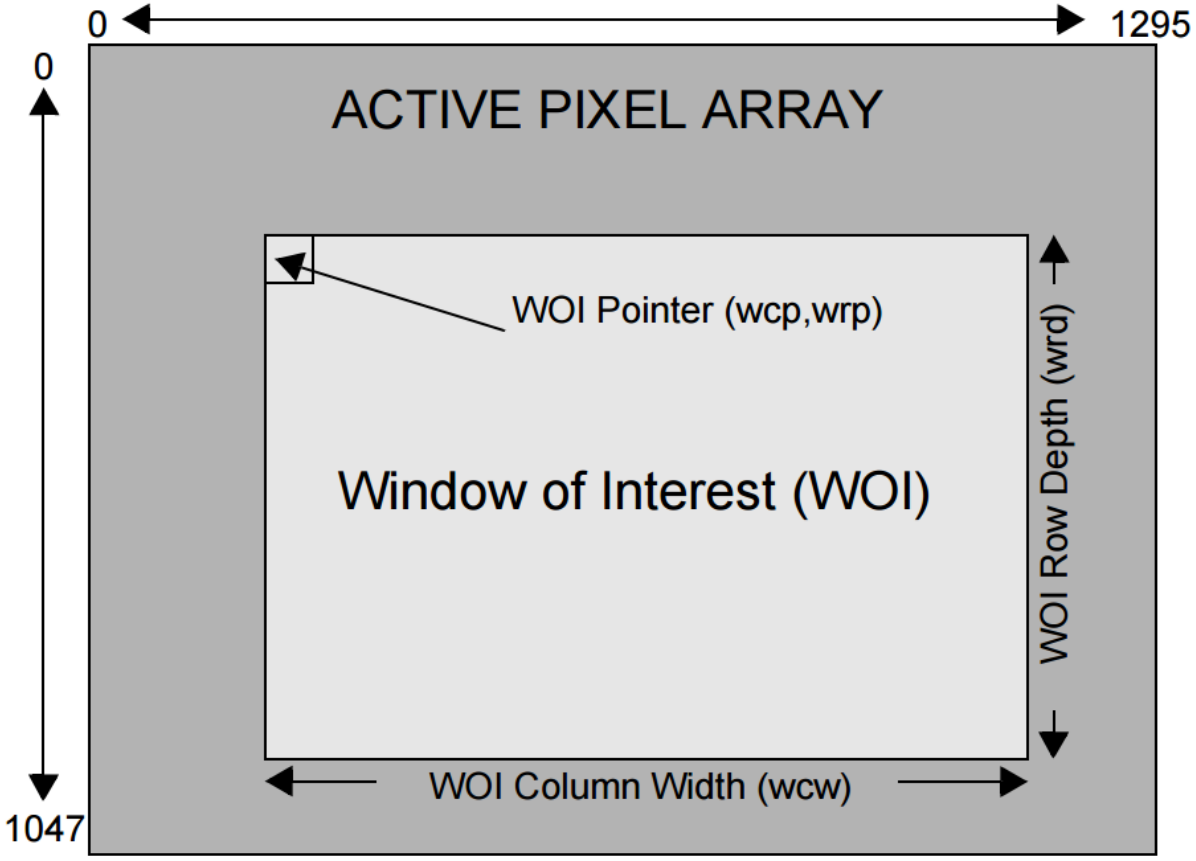
- **Parallel CMOS interface**
  Oldest, simplest interface using multiple data lines plus clock/synchronization signals; still found in lower-speed sensors

- **MIPI CSI-2**: *Mobile Industry Processor Interface – Camera Serial Interface 2*
  A high-speed serial interface originally developed for mobile devices, now widely used in embedded vision systems

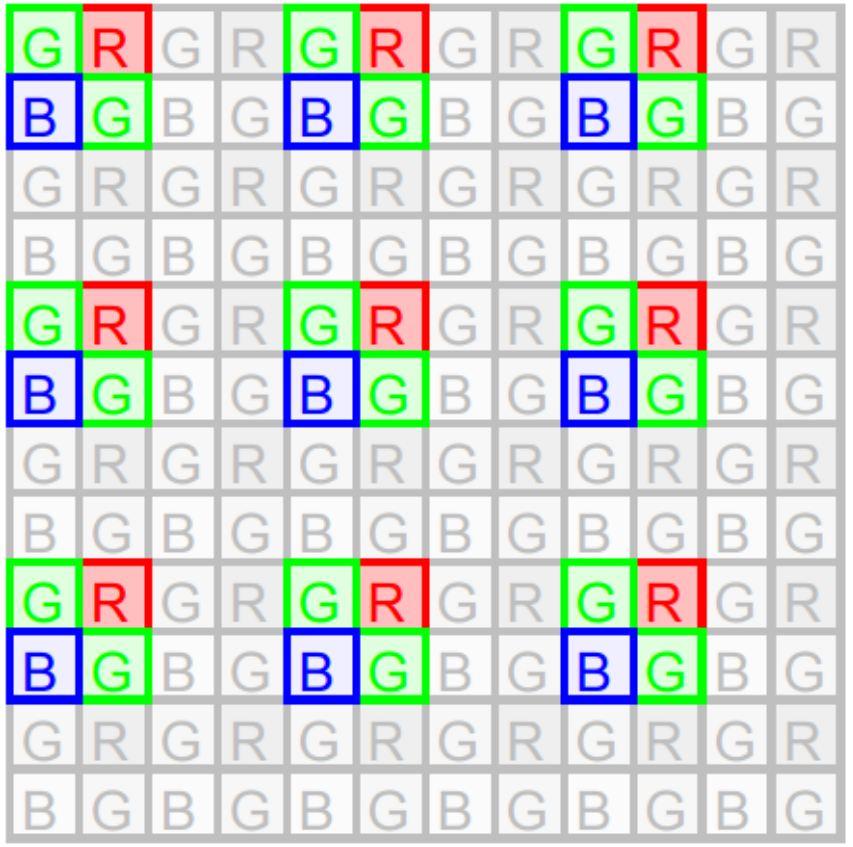
- **LVDS**: *Low-Voltage Differential Signaling*
  A differential electrical signaling standard that transmits data using a small voltage difference between two wires, providing good noise immunity at high speed

CMOS cameras typically communicates with PC through Camera Link, USB3 or Gigabit Ethernet

Window of interest



Window subsampling



→ speed up the readout

Rolling shutter

each frame is captured by scanning rows across the scene rapidly, so not all parts of the image of the scene are recorded at the same instant. This produces predictable distortions of fast-moving objects or rapid flashes of light.



Global shutter

the entire frame is captured at the same instant.

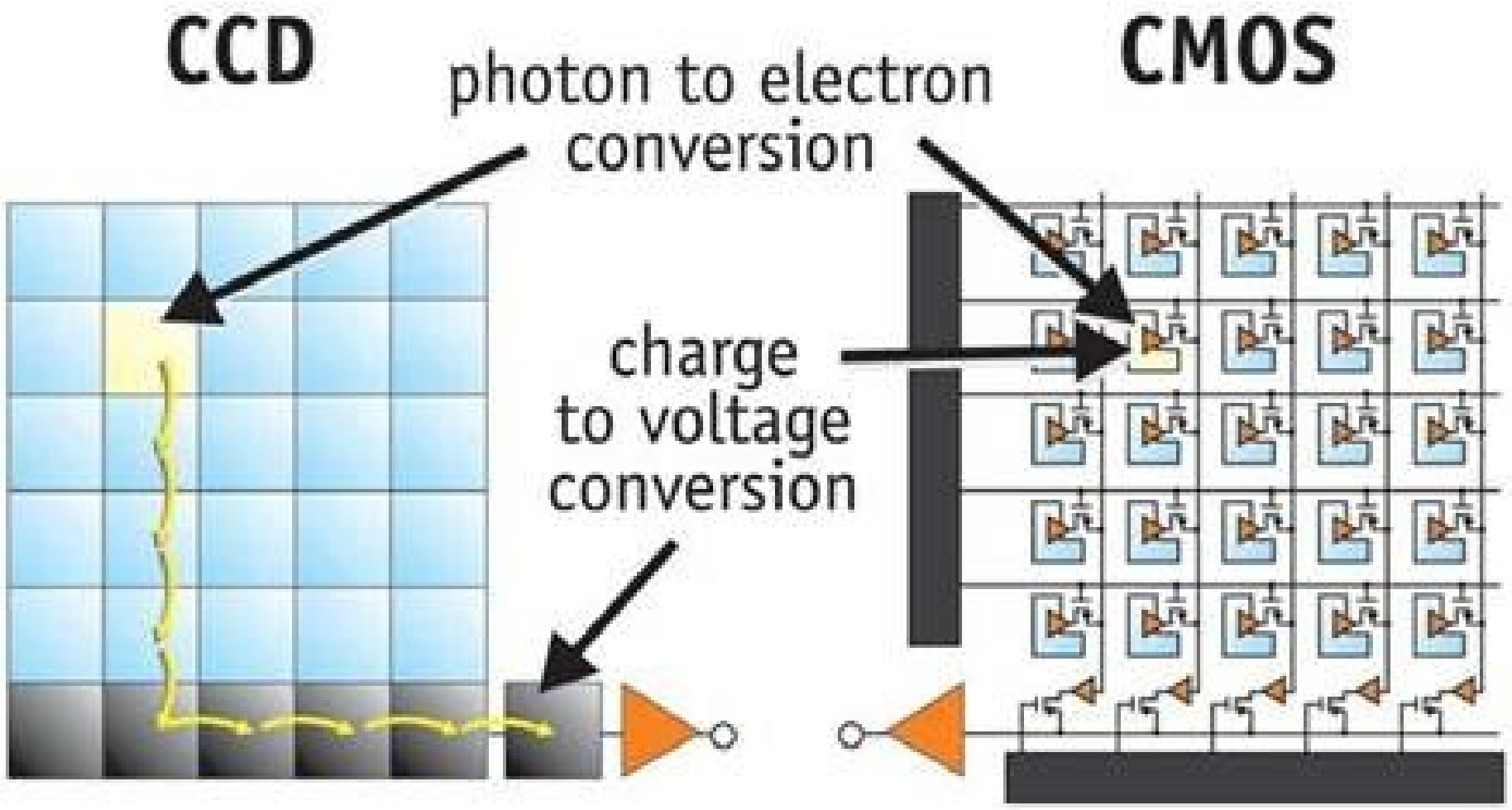


- High Volume Imagers for Consumer Applications (e.g., camera-phones)
- WebCams
- Imagers for Machine Vision (area and line scan)
- High Speed Motion Capture Cameras
- Digital Radiography
- Endoscopy (pill) Cameras
- DSLRs (Digital Single-Lens Reflex) camera








CCD
photon to electron conversion
CMOS
charge to voltage conversion

CCD advantages:

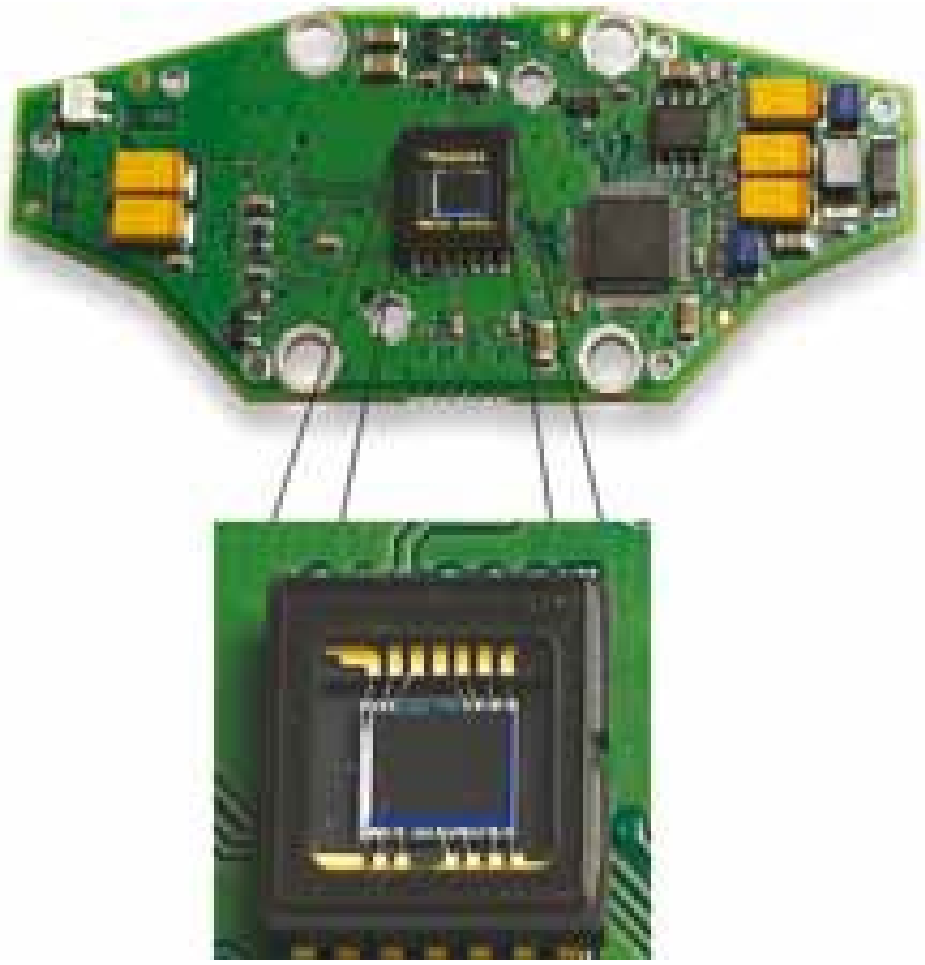
- Uniformity (less structured noise)
- Higher Fill-Factor
- Almost all chip area devoted to light capture
- Higher sensitivity in the Near Infra-Red (NIR)

CMOS advantages:

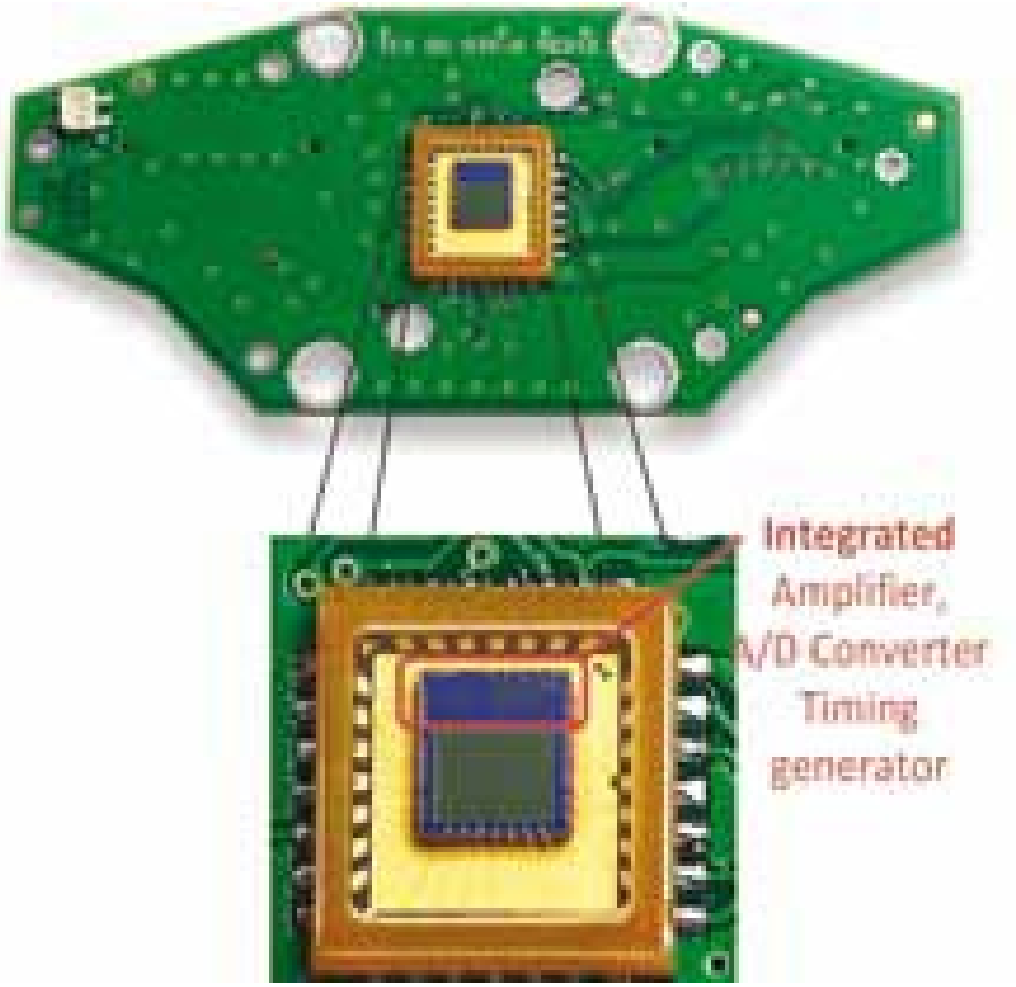
- Faster readout
  (one charge to voltage converter per pixel)
- Selectable active window (for higher frame rate)
- Digital output
- Less electronics outside sensor
  (internal Amplifier, ADC, processing…)
- Less power (→ lower temperature)

Cost considerations

- Custom imager → CCD
- Large scale economy → CMOS



CCD sensor



CMOS sensor